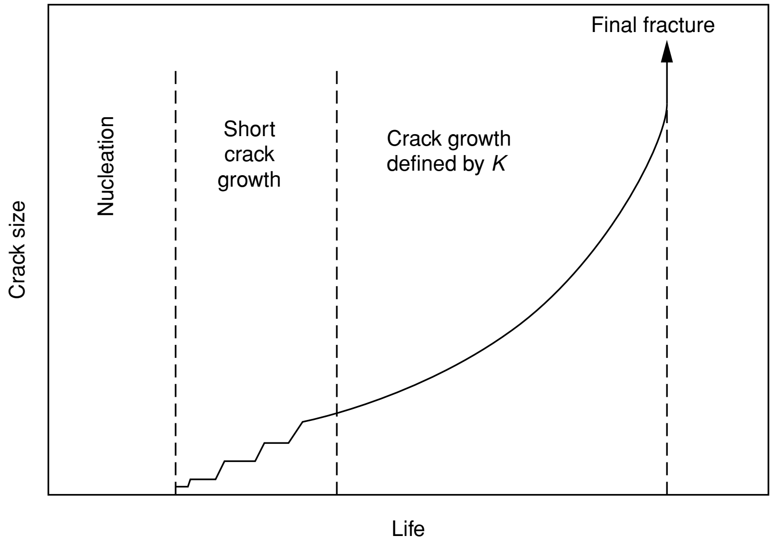

Fig. 4.42. Illustration of the different stages associated with fatigue crack failure: initiation, short crack growth, fatigue crack propagation and final fracture governed by the fracture toughness.

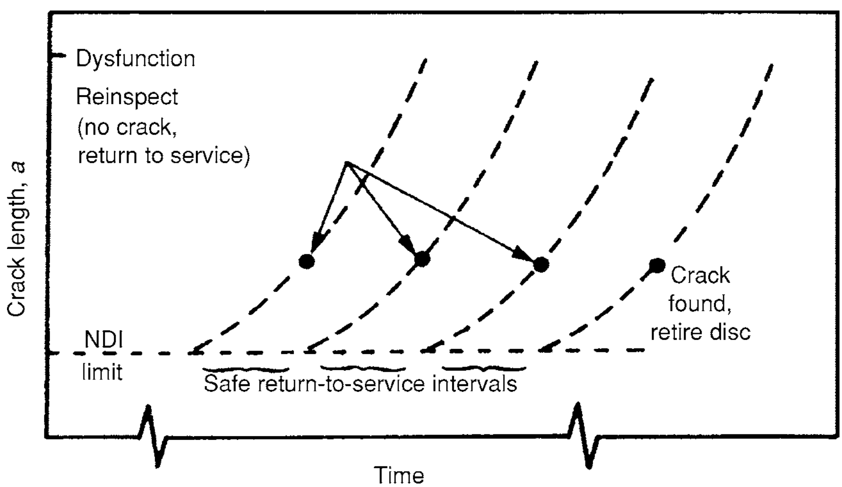

Fig. 4.43. Schematic representation of the damage-tolerant life prediction method [49].

the number of cycles required for fatigue crack growth to be initiated; see Figure 4.42. After the disc has operated for an appropriate number of cycles, it undergoes thorough inspection using the NDE techniques which would have been first employed before the disc entered service; provided that this reveals that no flaws are present, it is placed back into the engine for a further number of cycles. Figure 4.43 illustrates this so-called 'damage-tolerant' method [49].

It is helpful to make estimates of the safety margins to be expected from the use of nickel-based alloys of differing proof strengths, following the analysis presented in ref. [48].

Waspaloy exhibits a proof strength of about 800 MPa, Astroloy about 1000 MPa and Rene 95 about 1200 MPa; these values do not depend too strongly upon temperature below 600 °C. In each case, we use our estimates for K_{1c} and da/dN from before, with $\Delta\sigma$ placed equal to the proof strength. We determine the size of a_{NDI}, which must be detected if burst (or complete dysfunction) is not to occur before 10 000 cycles. The values of a_{NDI} and the crack lengths at one-third, two-thirds and complete dysfunction are tabulated in Table 4.2, from which some important conclusions can be drawn. For proof strengths of around 800 MPa, corresponding to an alloy such as Waspaloy, the crack reaches a size of 0.375 mm (deemed 'failure' in the life-to-first-crack approach) at only 1500 cycles, which is only 15% of the total number of cycles to failure. For Astroloy, it is reached at about 6300 cycles. However, for Rene 95, life-to-first crack is 8300 cycles, which is well beyond two-thirds of dysfunction. It is clear that the risk associated with the life-to-first-crack approach is greater when the alloy is stronger.

The computations presented here are obviously very simplistic and the various safety factors associated with the requirement to work at the -3σ quartile have not been accounted for. However, some important points emerge [48], as follows.

(i) Control of the manufacturing processes to prevent defects from being introduced is an absolute necessity. If pre-existing defects of length greater than 1 mm were to be present at service entry, catastrophic consequences might result.
(ii) The size of initial defect in the higher-strength alloys such that an adequate life of 10 000 cycles is reached is very small indeed. This serves to emphasise the need for very careful, non-destructive evaluation before entry into service.
(iii) The lower-strength alloys exhibit superior tolerance to defects, and consequently their reputation as fatigue-resistant materials is justified. It is clear why there can be some reluctance to replace these with the new alloys.

C The probabilistic approach to lifing

The dominant damage mechanism occurring in the superalloys used for turbine disc applications is crack initiation by low-cycle fatigue (LCF) and subsequent growth by cyclic fatigue crack propagation. For this reason, much emphasis has been placed on laboratory experimentation using test pieces which, for practical reasons, have relatively small dimensions: of perhaps ~10 mm diameter and ~50 mm gauge length. The accuracy of this approach requires that the conditions experienced by the test pieces are representative of those in a turbine disc. This is expected to be so when the density of nucleation sites is large, which will be the case for LCF conditions with high levels of cyclic strain or for superalloys of low yield stress such as Waspaloy or IN718. For these alloys, crack initiation is found to be associated with intense slip bands which have a very high density, and the damage is representative of that occurring in turbine discs of much greater size. The lifing approach is then essentially deterministic, although scatter in the laboratory test data should, of course, be accounted for.

The situation is somewhat different for alloys such as Rene 95 or RR1000, which have significantly higher strengths and which are manufactured by powder-processing routes.

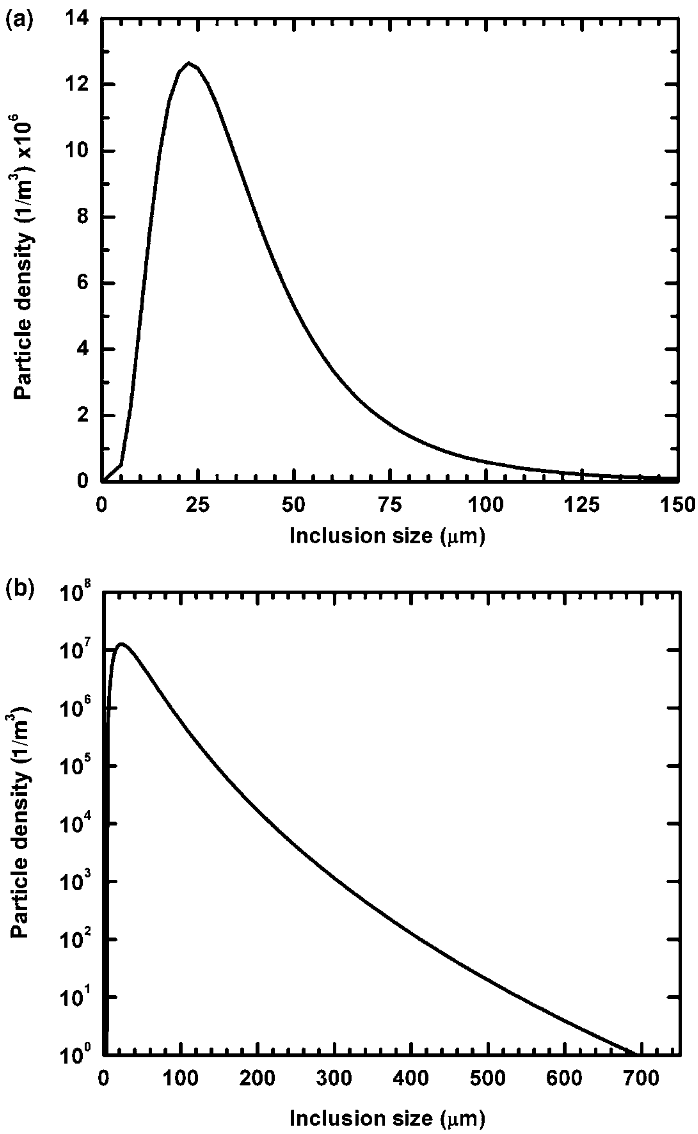

Fig. 4.44. Typical distribution of inclusions in a powder metallurgy alloy such as Rene 95 [50]. (a) Number of particles per unit volume against particle size, plotted on linear–linear axes; (b) as for (a), but plotted on log–linear axes.

First, the higher strength means that a higher proportion of the fatigue life tends to be spent in the fully elastic regime, in which the density of slip bands is low. Thus, testing conditions are not necessarily representative of the service environment, and a significant size effect might be anticipated. Second, failure is found to be associated with ceramic inclusions – typically Al_2O_3 or SiO_2 – which are inherited from the processing route. A typical particle size distribution for the Rene 95 superalloy [50] is given in Figure 4.44.

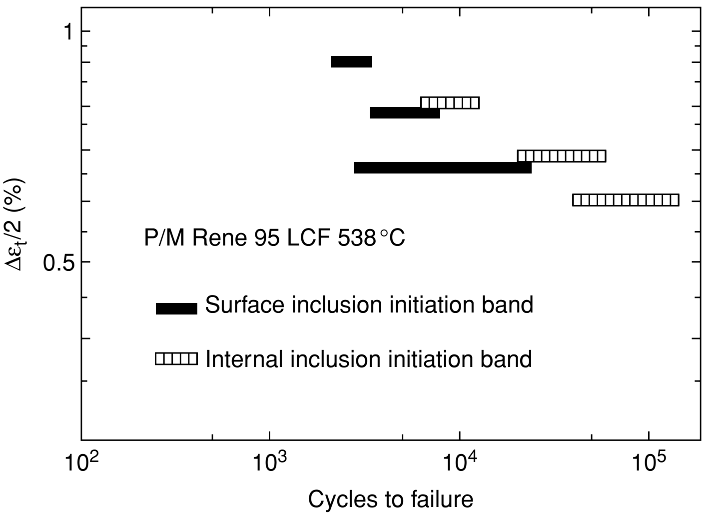

Fig. 4.45. Results from LCF fatigue tests on the powder metallurgy alloy Rene 95. Also indicated are the positions of the inclusions causing failure [51].

Although their density is very low – a fraction of less than 10^{-6} is typical – the frequency of their occurrence is sufficient to cause considerable scatter in the fatigue data. Figure 4.45 shows typical LCF data for Rene 95 [51]; note, in particular, the greater scatter at lower cyclic strains. Furthermore, in such tests it is found that inclusions at the free surfaces are very much more detrimental than inclusions initiating cracks at depth. These considerations emphasise the need for a statistical, or *probabilistic*, approach, as the fatigue performance now depends upon the surface area and volume of material being loaded. Larger volumes of material will perform more poorly than smaller ones, owing to the greater probability of an inclusion of greater size being present.

Pineau and co-workers have put these considerations onto a quantitative basis, and what follows is an analysis of the approach proposed in ref. [50]. Consider a volume, V, of material which has a surface area, S, exposed to a cyclic stress, $\Delta\sigma$, with $R = 0$. Assume that *uniform* spherical inclusions of diameter D are present at a density n_v per unit volume. The probability of any single inclusion avoiding the surface depends upon the surface to volume ratio, S/V; to a first approximation, it is given by the quantity $1 - DS/V$, if stereological effects associated with the intersection of the particle with the free surface are neglected. However, a total of $(n_v \times V)$ inclusions are present. It follows that the probability, p_1, of *any one* inclusion intersecting the free surface is given by

$$p_1\{D\} = 1 - \left(1 - \frac{DS}{V}\right)^{n_v V} \tag{4.10}$$

It is necessary also to acknowledge the manner in which any given inclusion interacts with the free surface; for example, a fully embedded inclusion is clearly more damaging than one which is only partially embedded to depth d, where $d < D$. If the flaw size, d, defines a fatigue life, N_0, consistent with the damage-tolerant approach described above,

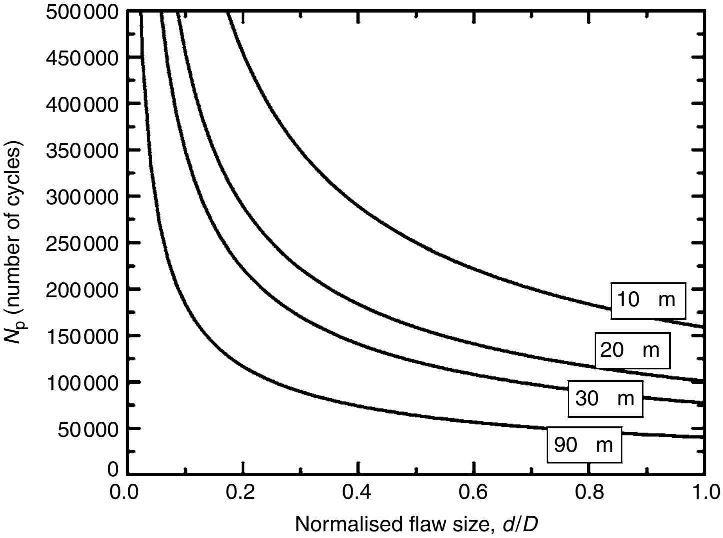

Fig. 4.46. Variation of the predicted life with the flaw size, d, normalised against the inclusion diameter, D, for various values of D.

the probability, p_2, that the *actual* life is N_p such that $N_p < N_0$ is

$$p_2\left\{N_p < N_0\right\} = \left[1 - \left(1 - \frac{DS}{V}\right)^{n_v V}\right]\left[\left(\frac{D-d}{D}\right)\right] \tag{4.11}$$

The term $(D-d)/D$ should be considered to be a 'harmfulness criterion', which acknowledges that, when $d = D$, the inclusion is at its most potent; then $N_p = N_0$ and $p_2 = 0$. At the other extreme, when the inclusion only just intersects the free surface, $d = 0$, and therefore it is not damaging; hence $p_2 = p_1$. In this limit, the N_p tends to infinity. Figure 4.46 illustrates the way in which N_p varies with d/D, for the various sizes of inclusion D implied by the particle size distribution in Figure 4.44. Note that as d/D tends to zero, an infinite life is expected since flaws are then absent.

In practice, a superalloy turbine disc material contains a *distribution* of particle sizes, and this must be accounted for. Suppose that the particle size distribution is discretised such that is contains n_{classes} classes, each of diameter D_i and number density n_v^i, such that $i = 1 \rightarrow n_{\text{classes}}$. The probability, p_3, that the life, N_p, is less than the potential life N_0 is then given by

$$p_3\{N_p < N_0\} = 1 - \prod_{i=1}^{n_{\text{classes}}} \left[1 - p_4\{N_p < N_0\}\right] \tag{4.12}$$

where, by appealing to the expression for p_2, one has

$$p_4\{N_p < N_0\} = \left[1 - \left(1 - \frac{D_i S}{V}\right)^{n_v^i V}\right]\left[\left(\frac{D_i - d}{D_i}\right)\right] \tag{4.13}$$

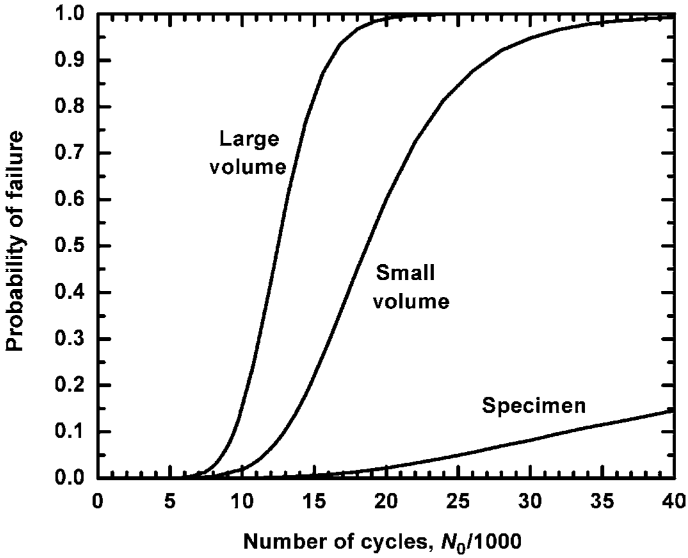

Fig. 4.47. Variation of the probability of failure, given by p_3, against the number of cycles, N_0, for the probabilistic model of Rene 95.

Note that the product in the expression for p_3 represents the probability that *every* class of inclusion, D_i, in the set of n_{classes} is non-damaging; p_3 is then the probability that *any one* of the D_i provides the fatal flaw.

With this model, quantitative calculations can be carried out in the following way. First, the variation of N_0 with flaw size, d, is determined using the damage-tolerant approach described in the previous section. Data for K_{1c}, $\Delta\sigma$ and the constants in the Paris law are required. Second, the data for d are normalised against the D_i present in the particle size distribution, and a family of curves is produced which describes the variation of the life, N_p, with d/D_i, as in Figure 4.46. Third, a prescribed life, N_0, is chosen; this defines d, and hence d/D_i, for any given D_i so that a set of values for p_4 can be determined from the expression given, each for a corresponding D_i. This allows a value for p_3 to be calculated. The process can be repeated for various values of N_0 and $\Delta\sigma$; see Figure 4.47, in which N_0 is plotted against p_3, the probability that the life falls short of N_0.

The results in Figure 4.47 – calculated for the Rene 95 superalloy with $\Delta\sigma = 1000$ MPa, $Y = 0.67$, the constants for the Paris equation given in the previous section and the particle size distribution given in Figure 4.44 – demonstrate that the fatigue performance displays a considerable dependence upon the surface area of the volume under load. For a large disc of surface area 84 000 mm^2 and volume 10^{-2} m, the probability of failure varies from less than 0.01 for fewer than 5000 cycles, to greater than 0.99 beyond 20 000 cycles. A smaller disc of surface area 9300 mm^2 and volume 3×10^{-3} m, or else a standard fatigue test specimen of surface area 200 mm^2 and volume 3.5×10^{-6} m, exhibit considerably superior performance. At any given probability of failure, the model indicates that the inclusion which is the likeliest to cause failure has a size which depends upon the surface to volume ratio. For example, a test specimen is predicted to fail from an inclusion at

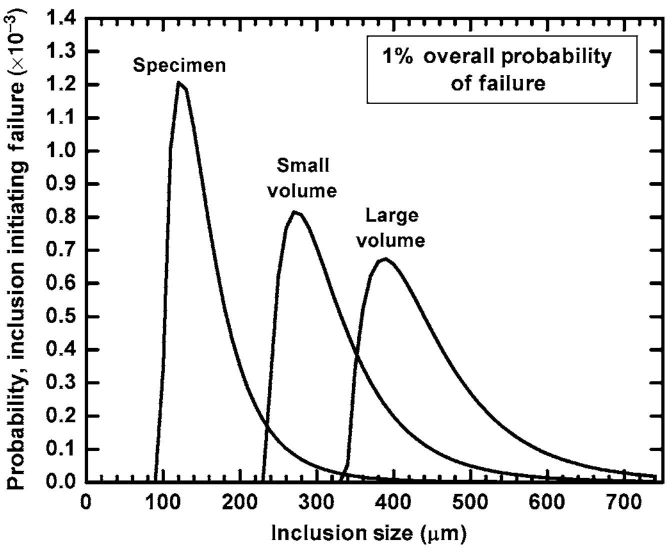

Fig. 4.48. Variation of the probability, p_4, with inclusion size, D_i, at a probability p_3 equal to 0.01, for the three geometries considered in the probabilistic model for Rene 95. The applied stress is 1000 MPa.

the lower end of the particle size distribution, whereas a turbine disc of reasonable size fails from an inclusion of greater size; see Figure 4.48. This is a direct consequence of a particle-sampling effect – a large particle (which has a low likelihood of occurrence) has a greater probability of causing failure when the surface area is significant. The sensitivity of the fatigue life to the particle size distribution has been confirmed [52] using low-cycle-fatigue (LCF) tests on powder-processed Udimet 720, and the results have been compared with those from the same material that had been intentionally doped with alumina particles of sizes 54 μm and 122 μm. The results (see Figure 4.49) indicate that the fatigue life is indeed impaired as the particle size increases; the reduction in life is greatest for lower total strain ranges and larger R-ratios. Figure 4.50 illustrates the results of fractography to identify the location of fatigue initiation in a typical turbine disc alloy. In principle, such experiments can be used to deduce the naturally occurring particle size distributions from the fatigue data of unseeded material – as required by the probabilistic modelling.

The above considerations justify the considerable emphasis which is placed by the superalloy industry on 'superclean melting technology' – the aim of which is to prevent non-metallic inclusions from becoming incorporated into superalloy forging stock and hence the turbine discs themselves. It is common nowadays for the rate of contamination to be less than one part per million on a weight per cent basis. Nevertheless, when using a probabilistic lifing method it is absolutely necessary to have a good estimate of the particle size distribution; the accuracy of the fatigue life depends upon it and, indeed, no estimate is possible if this is unavailable. To glean the necessary information, different approaches have been employed. Quantitative metallography has been used to examine plane, polished cross-sections. However, for 'superclean' material this can be problematical, since the number and

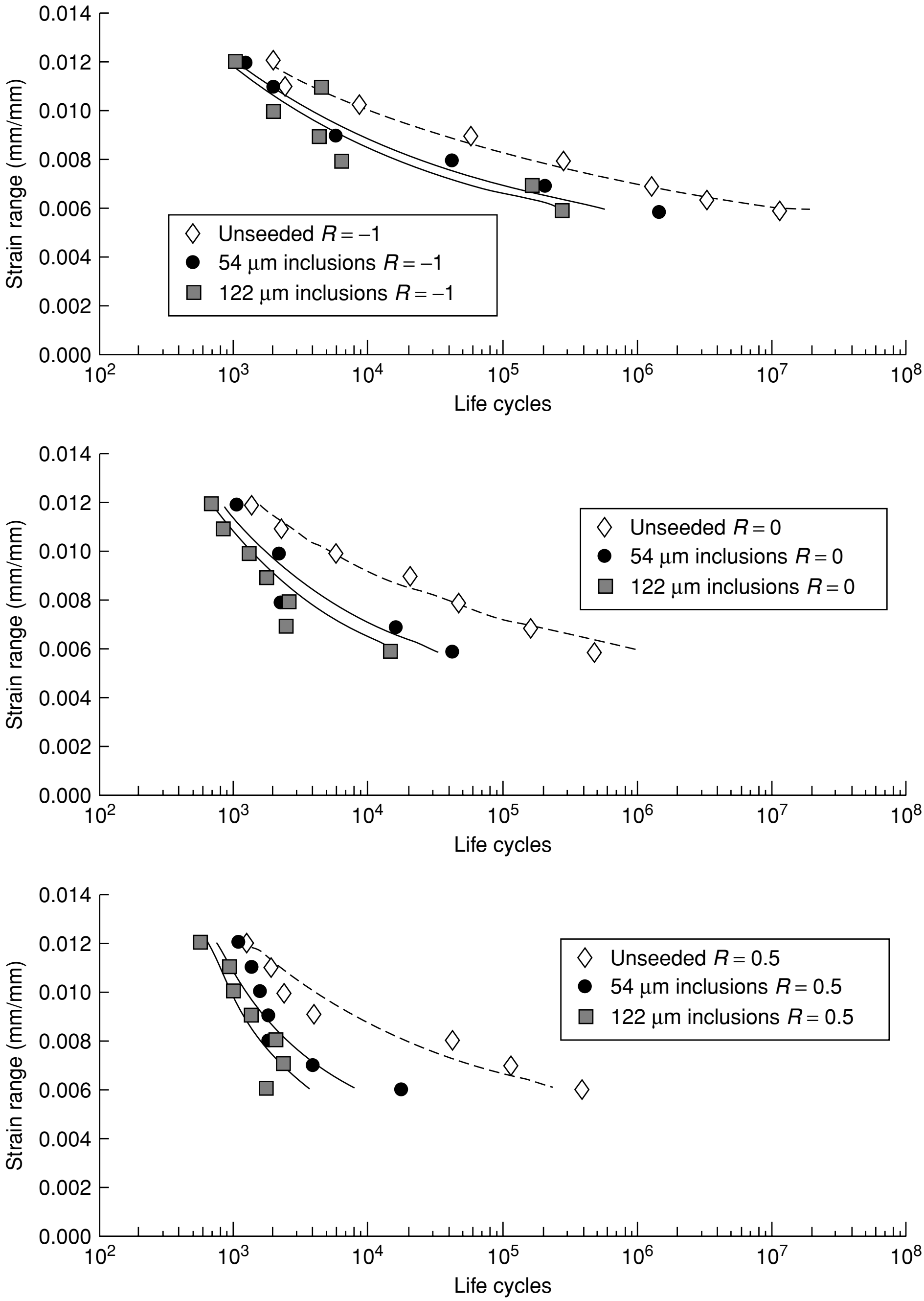

Fig. 4.49. Comparison of the LCF lives of powder metallurgy Udimet 720 in the unseeded and seeded conditions, for various R ratios [52].

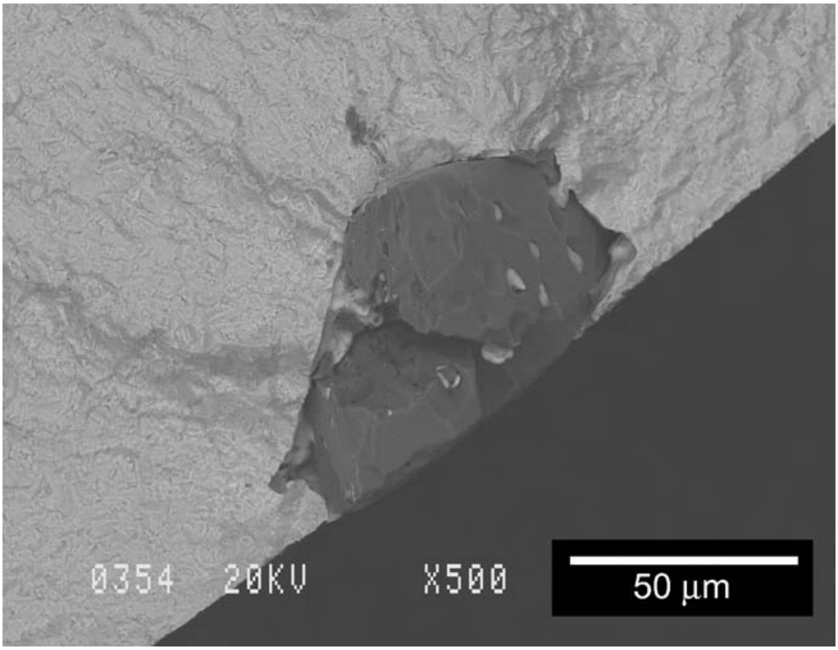

Fig. 4.50. Fatigue initiation site for powder metallurgy Udimet 720 seeded with 54 µm diameter inclusions, LCF tested at 427 °C under strain control with a range of 1.2%, $R = -1$ [52]. Surface initiation is apparent.

size of inclusions is very small, possibly to an extent which makes the work prohibitively time-consuming and expensive. This is confirmed by the data in Figure 4.51, which shows the cross-sectional area of material to be examined in order to obtain an estimate of the volume fraction of inclusions to an accuracy greater than 20%, with 80% confidence [53]; for a diameter of 100 µm and a volume fraction of 10^{-6}, the surface area required is $\sim 10^6$ mm^2, or one square metre. These difficulties have provided the incentive for the development of the electron beam button melting (EBBM) technique, in which about 1 kg of the alloy is drip-melted under vacuum by an electron beam; the inclusions rise to the surface of the pool (since they are lighter than the molten liquid), where they are subject to buoyancy and Marangoni forces. If the cooling is then properly controlled, the inclusions concentrate into the last portion to solidify, which then allows them to be analysed. The EBBM techique has shown some promise, although it has been criticised on the grounds that it does not provide an accurate measure of the *distribution* of particle sizes, since the flotation efficiency is a function of the particle size.

4.3.3 Non-destructive evaluation of turbine discs

It has been shown in Section 4.3.2 that the flaws in a turbine disc are of great significance; since they represent potential sites for the nucleation of cracks, they threaten the integrity of the component and reduce the fatigue life. For this reason, much emphasis is

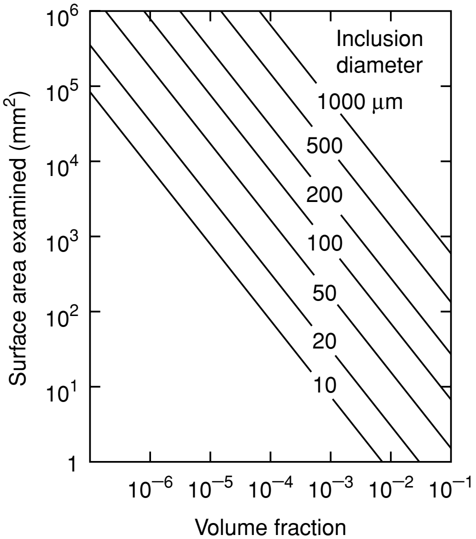

Fig. 4.51. Variation of the surface area which must be examined with volume fraction and diameter of inclusions, at the 80% confidence limit [53].

placed on using non-destructive evaluation (NDE) to characterise the dimensions of the flaws, both during the manufacturing of the turbine disc – i.e. before the turbine disc enters service for the first time – to ensure that it is of acceptable quality, and then whenever the disc is removed from the engine, prior to a decision being made to place it back in service. Obviously, the first situation arises regardless of the lifing method employed, the second only when the damage-tolerant method is being used. The various NDE methods which are suitable include (i) the liquid penetrant method, most usually using dyes which fluoresce under ultra-violet light, thus providing a measure of the length of each crack exposed to the surface; (ii) the eddy current method, which enables a quantitative estimate of the crack depth, but only for cracks located at or near the surface; (iii) X-ray radiography, which has the power to penetrate relatively thick sections, but which cannot usually resolve the finest flaws that are present; and (iv) ultrasonics – the use of high-frequency sound waves for the detection of inclusions, coarse grains, shrinkage pipe, tears, seams and laps. In practice, for turbine disc applications this final method is of the greatest importance.

To illustrate the use of ultrasonics [54] for NDE in this context, consider its application to a forged and heat-treated disc, which will have been machined to a pre-specified geometry known as the 'sonic shape' or 'condition of supply'. To enable ease of ultrasonic inspection, the features of the disc (for example, the blade root sockets, cover plate and flanges) are not introduced by the machining process as yet, so that, at this stage, it contains only a few, flat surfaces which intersect at 90° angles; see Figure 4.52. To detect the smallest flaws, care is taken also with the surface finish, which is typically better than 6 μm. Production and measurement of the ultrasonic waves is accomplished via the use of piezoelectric transducers and receivers – these rely upon crystals, such as quartz or barium titanate, which display a strong inverse piezoelectric effect. Since ultrasonic waves are attenuated strongly by air, the

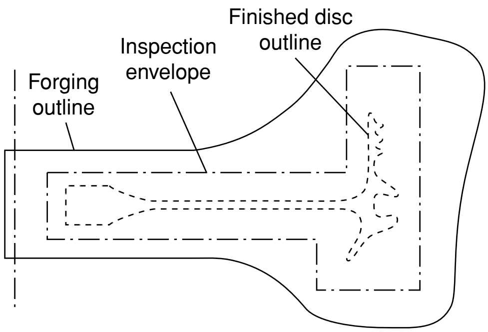

Fig. 4.52. Schematic illustration of the outline of a machined turbine disc, the sonic envelope used to test the material ultrasonically and the forging from which it is machined.

disc and transducer are either kept immersed in water during inspection, or else a thin layer of coupling fluid is maintained between the two; the waves are attenuated only very weakly by the material itself, so that very thick sections can be analysed. In practice, the waves are sent through the component in at least two different ways. First, the so-called 'straight-beam top inspection' is carried out to detect planar flaws lying parallel to surfaces whose normal lies parallel to the axis of the forging; for this purpose, a longitudinal ultrasonic beam is directed parallel to the normal to these same surfaces, using a single pulse-echo transducer, 13–40 mm in diameter, at a frequency in the range 1–5 MHz. Calibration is carried out using standard 50 mm diameter cylindrical reference blocks of varying heights, into which flat-bottomed holes are machined to simulate artificial flaws of different sizes located at different depths. Detection sensitivity is normally set to 10% of the size of the flaw machined into the calibration block. For example, if the #2 block in Series B of the ASTM standard E127 is found to be suitable for calibration [55], the expected sensitivity of flaw detection is then 0.08 mm since the flat-bottomed hole in that case has a diameter of 0.8 mm. A second set of tests are conducted using shear waves, to detect flaws that have an axial radial orientation; for this, an angle-beam transducer is used to introduce a beam at 45° to the surface normal, and inspection is performed by scanning in the circumferential direction, both clockwise and anti-clockwise, around the periphery of the forging. Calibration notches – typically 25 mm long and either V-shaped or rectangular – are cut axially on the inner and outer surfaces of the forging, with a width not exceeding twice the depth. The sensitivity of the detection is established by first adjusting the instrument controls to obtain a minimum 13 mm sweep-to-peak signal from the calibration notch on the outer surface, followed by a measurement of the response from the notch on the inside surface; the peaks corresponding to the two notches are connected thus establishing a reference line.

With regard to the inspection of different superalloys, the background noise is found to be significantly greater for coarse-grained material, and therefore the sensitivity of flaw detection is then much reduced; small flaws are then more likely to be undetected. The data given in Table 4.5 illustrate this [56]. Interestingly, for any given grain size, powder metallurgy (P/M) material exhibits ultrasonic inspectability which is very much better than for conventional cast-and-wrought product. This has been attributed to the

Table 4.5. *Data for the inspectability of the Udimet 720 superalloy for various grain sizes, in the powder and cast-and-wrought conditions [56]*

Form	Billet diameter /mm	Grain diameter /μm	Noise level
Powder metallurgy, extruded	150–300	3	6% at hole 1[a]
Powder metallurgy, sub-solidus HIP + cog	300	11	25% at hole 1
Powder metallurgy, HIP	165	11	15% at hole 1
Powder metallurgy, sub-solidus HIP	300	127	50% at hole 1
Cast and wrought	165	8	100% at hole 2

Note: [a] Hole 1 corresponds to a diameter of 0.40 mm.

more uniform grain size and the reduced levels of segregation when powder metallurgy is employed.

Questions

4.1 Refining of superalloys is often done with a combination of vacuum arc melting (VIM), electro-slag remelting (ESR) and vacuum arc remelting (VAR). Triple-melted stock is then processed by the combination of VIM/ESR/VAR, in that order. Explain why the induction-melted stock is subjected to electro-slag remelting prior to vacuum arc remelting, rather than the other way around.

According to Maurer, the maximum ingot diameter which can be supplied in the alloy IN718 to aerospace quality depends upon the secondary melting processes employed. For example, 17 inch is the limit for a VIM/ESR ingot, 20 inch for a VIM/VAR ingot and 24 inch for a triple-melted VIM/ESR/VAR ingot. Explain why this should be the case.

4.2 Careful control of electrode immersion is important during ESR refining, with a 1/4 inch immersion of the electrode being suggested for optimum processing. Why is control of the immersion depth important? What would be the consequences of (i) too great an immersion depth and (ii) too small a depth? What would be the influence on the voltage swing?

4.3 The ESR process gives rise to considerably less shrinkage pipe than the VAR process. Why is this?

4.4 During the secondary remelting of IN718 superalloy using the VAR process, freckling can occur. The most usual location is at mid-radius, implying that the material at centre and surface positions is not so freckle-prone. Explain.

4.5 During the solidification of the superalloys, the formation of TiN particles ahead of the growing solid/liquid interface has been implicated in the nucleation of stray grains. Explain how you would expect this columnar-to-equiaxed transition to depend upon (i) the growth velocity and (ii) the temperature gradient employed.

4.6 Cast ingots of the alloy IN718 are found to contain significant quantities of the Laves phase, $(Ni, Fe)_2Nb$. This can be found on the binary Fe–Nb phase diagram. This is despite the fact that IN718 contains only 5.35 wt% Nb. What is the reason for this? Would you expect powder IN718 to have more or less of the Laves phase?

4.7 Why, before the consolidation step in the processing of superalloys by powder-processing, is it usual to blend together particle distributions of different sizes?

4.8 Work aimed at optimising the compositions of polycrystalline superalloys for turbine disc applications has confirmed that additions of boron have a very beneficial effect on creep performance. Calculate the concentration of boron (in wt%) required to deposit a monolayer of boron on the grain boundaries of a superalloy with a grain size of (a) 1 μm, (b) 10 μm and (c) 100 μm. Do the concentrations calculated compare well with the values used in practice?

4.9 The microstructure of a polycrystalline turbine disc alloy can display the γ' phase in primary, secondary and tertiary forms. Give an explanation for this, invoking a description of possible heat-treatment schedules and the microstructural response expected due to nucleation, growth and coarsening theories.

4.10 A closed-die forging operation on a piece of a superalloy billet leaves a characteristic grain size distribution, which depends upon the thermal and strain history induced. Describe what this might look like. Will the grain size be greater at the rim or the bore region?

Explain why this grain size distribution is unfortunate, given the operating conditions experienced by the rim and bore in the gas turbine engine. How might a dual-heat-treatment process help in this regard?

4.11 Examine the low-cycle-fatigue data for various turbine disc alloys, which are given in Figure 4.30(a). At low total strain ranges, $\Delta\epsilon_t$, stronger alloys such as Rene 95 (each possessing a significant yield stress) have better fatigue lives than weaker ones such as Waspaloy. However, for large $\Delta\epsilon_t$, the converse is true. Give an explanation for this behaviour.

4.12 With regard to the lifing of a turbine disc, distinguish between the (a) life-to-first-crack and (b) damage-tolerant approaches to lifing. Which method would be the more appropriate for (i) a fatigue-resistant alloy such as that used for civil engines and (ii) a higher-strength alloy used for a military application?

4.13 The stress distribution in a rotating circular disc is of practical importance in the design of gas turbine engines. For a disc containing a central hole of radius a, the principal stresses in the plane of the disc depend upon its density, ρ, angular velocity, ω, and maximum radius, b, according to

$$\sigma_r = \frac{3+\nu}{8}\rho\omega^2\left(b^2 + a^2 - \frac{a^2b^2}{r^2} - r^2\right)$$

$$\sigma_\theta = \frac{3+\nu}{8}\rho\omega^2\left(b^2 + a^2 + \frac{a^2b^2}{r^2} - \left[\frac{1+3\nu}{3+\nu}\right]r^2\right)$$

For a disc containing a central hole with radius $a = b/10$, plot out the variation of σ_r and σ_θ with r. Take $\nu = 1/3$. Where are the maximum radial and hoop stresses? Compare the maximum value of the hoop stress with the maximum value for a disc with no hole ($a = 0$) and note the stress-concentrating effect of the hole. Hence explain why it is possible to spin a typical helicopter turbine disc very much faster than one in a civil aeroengine.

4.14 Derive an expression for the average stress, P_b, exerted by the blades on the rim of the turbine disc, which is taken to have a thickness, t, in the axial direction. Model each blade as a simple slab. Your expression should include the angular velocity, ω, the number of blades, N_b, the mass of each blade, M_b, and the typical dimensions. If the rotational speed of the HP shaft is 10 000 rev/min, estimate a value for P_b.

4.15 Lame's equation for the hoop stress at the bore of a penny-shaped disc of outer radius y and hole of radius x, loaded evenly around its outer circumference with a stress P_b, is given by

$$\sigma_{\text{hoop,bore}} = \left(\frac{1-\nu}{4}\right) x^2 \rho \omega^2 + \left(\frac{3+\nu}{4}\right) y^2 \rho \omega^2 + \frac{2P_b y^2}{(y^2 - x^2)}$$

where ρ is the density and ν is Poisson's ratio. Use this expression to make a first estimate of the inner and outer diameters of the turbine disc of a large turbofan such as the GE90. How does (i) the weight and (ii) the hoop stress, $\sigma_{\text{hoop,bore}}$, vary as x and y are altered?

4.16 The operating conditions of a civil aeroengine are at their most extreme at take-off, when the shaft speed increases typically by about 5% over the value in steady-state flight at altitude. Using the equations given in previous questions, compare the stress distributions in the HP turbine disc (i) at take-off and (ii) during steady-state flight. Why might your considerations be overly simplistic?

4.17 Four turbine discs are tested in a spin-pit test rig, under conditions which are representative of those seen in service. Failure occurs at 8821, 9276, 9946 and 10 592 cycles. On the basis of the *life-to-first-crack* lifing philosophy, predict a safe working life for this component.

After entry to service, one turbine disc is removed prematurely from the jet engine after 3000 cycles. The test rig is then used to spin it to destruction – this takes a further 5967 cycles. Given this information, calculate a revised predicted safe working life. State your assumptions. Has the *mean* life increased or decreased?

4.18 A gas turbine manufacturer is considering the use of Rene 95 for a high-pressure turbine disc, which is currently manufactured from Waspaloy. This will allow the range of cyclic stress, $\Delta\sigma$, to be increased from 800 MPa to 1200 MPa to reflect the increased yield stress of the new alloy. The fracture mechanics 'damage-tolerant' approach is used for turbine disc lifing.

With regard to the limits set for surface flaw detection using non-destructive inspection, what are the ramifications of specifying Rene 95 for this application? Assume that fatigue crack growth in both alloys can be described by

$$\frac{\mathrm{d}a}{\mathrm{d}N} = 4 \times 10^{-12} \Delta K^{3.3}$$

where ΔK is expressed in $\mathrm{MPa\,m^{1/2}}$ and $\mathrm{d}a/\mathrm{d}N$ is the growth increment per cycle expressed in metres, and that $K_{1c} = 100\,\mathrm{MPa\,m^{1/2}}$ for most superalloys. Take $Y = 0.67$, corresponding to a semi-circular edge crack.

4.19 At ambient temperature, the fatigue crack propagation rate of many polycrystalline superalloys is found to depend upon the ratio $\Delta K/E$, where ΔK is cyclic stress intensity and E is Young's modulus. Relationships such as

$$\frac{da}{dN} \propto (\Delta K/E)^{3.5}$$

have been proposed [57], where it has been shown that growth rates are predicted within a factor of ± 2 provided that a suitable calibration constant is chosen.

How might a normalisation of ΔK by E be justified? Suggest a better normalisation, involving the product of the yield stress and modulus, based upon the general principles of fracture mechanics. Are there circumstances under which these approximations might be too crude?

4.20 In practice, the processing of a high-strength turbine disc superalloy such as Rene 95 superalloy cannot be achieved without introducing distributions of ceramic inclusions; see Figure 4.44.

Determine the *most likely* diameter of surface-breaking inclusion in the Rene 95 superalloy, given this distribution in Figure 4.44, for (i) a specimen of surface area 200 mm^2 and volume 3.5×10^{-6} m^3, (ii) a helicopter disc of surface area 9300 mm^2 and volume 2.9×10^{-4} m^3, and (iii) a disc for a civil turbofan engine of surface area 84 000 mm^2 and volume 1.0×10^{-2} m^3.

Given the fatigue performance and loading situations outlined in the previous question, and assuming that fatigue crack growth begins at the first cycle from a flaw of size equal to the diameter of the most likely surface-breaking inclusion, estimate the fatigue life for each of the situations (i), (ii) and (iii) described above. Hence demonstrate that the predicted fatigue life has a considerable dependence on component size.

4.21 The turbine disc material of an engine is to be replaced with a stronger grade of alloy. This will allow the engine to be run harder, although care will be taken to ensure that the hoop stresses at the bore do not exceed the same proportion of the yield stress. How would one expect the (i) fatigue initiation and (ii) the fatigue propagation lives to be altered? Does this have ramifications for the way in which one must life the turbine disc?

4.22 Consider the *damage-tolerant* (sometimes known as the *retirement for cause*) method of lifing a turbine disc. In practice, the number of cycles declared after inspection is assumed not to vary during repeated use of the disc – i.e. it is the same for the first and last service overhauls. Explain why this is not an optimum strategy in some ways.

4.23 You are promoted to Chief of Materials for a company specialising in the design and manufacture of large civil aeroengines. Unfortunately, your organisation has no experience with the use of powder metallurgy product for its turbine discs. Prepare a two-page memorandum for the Board of Directors, which summarises the challenges which need to be overcome before powder metallurgy product can be introduced into the product line.

4.24 During the inspection of forged superalloy billet using ultrasonic non-destructive testing, the probability of successfully detecting a defect depends strongly upon the grain size of the material. Why is this? And why does this give powder billet a big advantage in this respect?

4.25 The magnetic particle inspection (MPI) method is used as a non-destructive evaluation technique for the detection of flaws in combustor cans, but it is used much more rarely for turbine discs. Why is this? Can you suggest an alloy which is likely to respond favourably to MPI?

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5 Environmental degradation: the role of coatings

As with any material, the superalloys suffer chemical and mechanical degradation when the operating temperatures are too high. Obviously the incipient melting temperature of a superalloy represents an upper limit on the temperature that can be withstood; this is usually no greater than about 1600 K. Despite this, the turbine entry temperature (TET) of the modern gas turbine continues to increase, with a take-off value of 1750 K being typical at the turn of this century; see Figure 1.5. Such extreme operating conditions have become possible only because action is taken to protect the components using surface engineering. In fact, the provision of such coatings and measures to ensure that they remain in place during service has become the most critical issue in the gas turbine field; in a state-of-the-art engine the components in the combustor and turbine sections would degrade very quickly were it not for the protection afforded by the coatings placed on them [1]. Thus, whilst the primary role of the superalloy substrate is to bear the mechanical stresses developed, an additional requirement is for mechanical and chemical compatibility with the coatings required to protect them.

Figure 5.1 summarises the different coating technologies which have become available [2,3] and ranks coating life and the temperature enhancement conferred by them in a relative way. It also serves as an introduction to the terminology used. The so-called diffusion coatings remain the most common form of surface protection. For example, aluminium is often deposited onto the surface of the superalloys by chemical vapour deposition, which, in its crudest form, is known as ‘pack-aluminisation’; subsequent heat treatment to promote adhesion encourages interdiffusion with the superalloy substrate such that an aluminium-rich layer is formed on the surface of the component – this is typically rich in the β–NiAl phase [4]. Due to their high surface concentration of aluminium, coated superalloys produced in this way are very efficient formers of an external, protective alumina layer or scale, and consequently resistance to oxidation is much improved. It has been demonstrated that the electrodeposition of a 5 to 10 μm-thick layer of platinum prior to the aluminisation process improves both high-temperature oxidation and hot corrosion resistance (for example, see ref. [5]). When treated in this way, superalloys are said to be protected by aluminide or platinum-aluminide coatings; indeed the platinum aluminide coatings have become an industrial standard against which other coating solutions are compared. For greater resistance to oxidation and corrosion, so-called overlay coatings are available, but at greater cost, since deposition must be carried out by air or vacuum plasma spraying (APS/VPS) [6] or else electron beam physical vapour deposition (EB-PVD) [7]. The MCrAlX-type materials in various compositional variants have become the standard overlay coatings for gas turbine

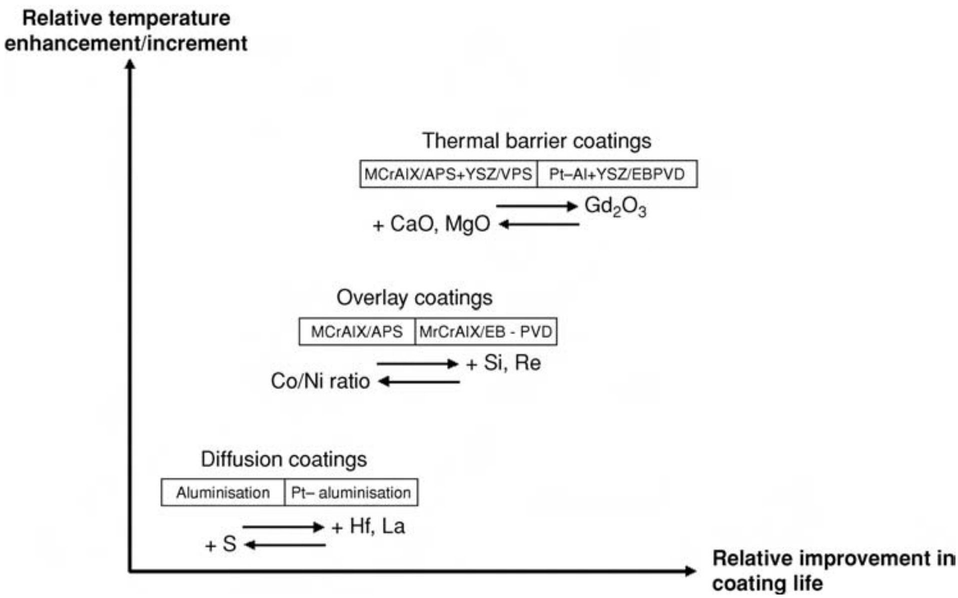


Fig. 5.1. Schematic illustration of the three common forms of protective coating used for turbine applications, and their relative coating lives and temperature enhancements which they afford.

applications [8,9]. Here, the 'M' refers to Ni or Co or combinations of these, and 'X' is a 'reactive' element (or mixture of them) at concentrations approaching 0.5 wt% – yttrium, hafnium and silicon are commonly used. The overlays have the advantage over the diffusion coatings of a greater flexibility of coating composition, with chemical and mechanical behaviour largely independent of the substrate on which they are placed; diffusion coatings on the other hand – by the nature of their formation – inherit a strong dependence on the substrate composition.

The thermal barrier coatings (TBCs) represent an alternative strategy – these are now finding very widespread usage in turbine engines [10]. A ceramic layer is deposited onto the superalloy, which, by virtue of its low thermal conductivity, provides thermal insulation and thus lowers the temperature of the metallic substrate. The effect can be very potent – a modern TBC of thickness 300 μm, if used in conjunction with a hollow component and cooling air, has the potential to lower metal surface temperatures by a few hundred degrees [11]. The first thermal barrier coatings emerged in the 1960s and were produced by the plasma spraying of calcia- or magnesia-stabilised zirconia. Whilst these were observed to perform well below $\sim$1000 °C, above this temperature they were found to be unstable; formation of MgO and CaO occurred by diffusion of Mg^{2+} and Ca^{2+} ions, and the formation of the monoclinic form of zirconia was promoted causing a four-fold increase in the thermal conductivity. Modern TBCs that are now widely used to protect turbine blading, nozzle guide vanes and combustor sections in service are based upon zirconia containing about 7 wt% of yttria – so-called yttria-stabilised zirconia (YSZ) – a composition identified originally by NASA as giving the best thermal cycle life in burner rig tests. Unfortunately, TBCs based upon YSZ are unable to prevent ingress of oxygen due to rapid transport of oxide ions through them and, consequently, oxidation of the underlying substrate remains a possibility during operation. This fact, along with the mismatch in thermal expansion coefficients of superalloy substrate (~17 ppm/°C) and ceramic top coat (~12 ppm/°C), which causes the formation of thermal stresses during thermal cycling, leads to the eventual failure of the TBC during operation by spallation. To mitigate against this possibility, a so-called 'bond coat' is deposited on the superalloy substrate prior to the deposition of the ceramic top coat; some form of diffusion or overlay coating is used for this purpose [12,13]. The term TBC therefore strictly refers to the combination of ceramic layer and bond coat; these need to be chosen to ensure compatibility with the superalloy substrate. Even with a bond coat in place, spallation of the ceramic layer will eventually occur, albeit after a more prolonged period of time. Promoting TBC life and the design of procedures to estimate it are currently among the biggest challenges faced by materials scientists and engineers working in the gas turbine field [14].

5.1 Processes for the deposition of coatings on the superalloys

5.1.1 Electron beam physical vapour deposition

The electron beam physical vapour deposition (EB-PVD) method for the coating of gas turbine materials was introduced by Pratt & Whitney in the late 1960s, for the production of overlay coatings of the MCrAlX type [7]. Whilst the method is still used widely today for this purpose, the EB-PVD process has come into its own for the deposition of the ceramic TBC materials for turbine blade aerofoils [10]; for the overlays, EB-PVD has assumed decreased importance because many of these are now processed using plasma spraying (see Section 5.1.2). The characteristic feature of the EB-PVD process is the use of an electron beam to vaporise an ingot of the coating material, which is typically held in a water-cooled copper crucible; this causes a vapour cloud to be formed above the ingot in which the component for coating is manipulated; see Figure 5.2 [15]. As the coating material is evaporated, the feedstock is fed into the chamber to maintain a constant feedstock height – it is important to generate a completely molten evaporant surface free of splashing so that the coating process is stable. Thus no chemical reactions are involved. In order to increase the productivity of the process to a reasonable level (coating rates of between 5 and 10 μm per minute are typical), a powerful electron beam gun needs to be employed, for example, a 150 kW unit operating at 40 kV. Moreover, processing is carried out in a vacuum of better than a few Pa. Since there is also a requirement for the manipulation of the components within the vapour cloud, it turns out that the EB-PVD process is one of the most expensive of the coating processes used in the gas turbine industry. In practice, there are only around 50 establishments around the world which are capable of depositing EB-PVD TBCs to the required standard for engine service. Figure 5.3 shows an industrial-scale EB-PVD unit used for the deposition of coatings on turbine blading.

It has been found that superior TBC density, hardness, erosion resistance and spallation life are achieved only when the substrate temperature is raised to within the range 850 °C to 1050 °C during processing. By design, the resulting morphology then consists of a series of columnar colonies which grow competitively in a direction perpendicular to the surface

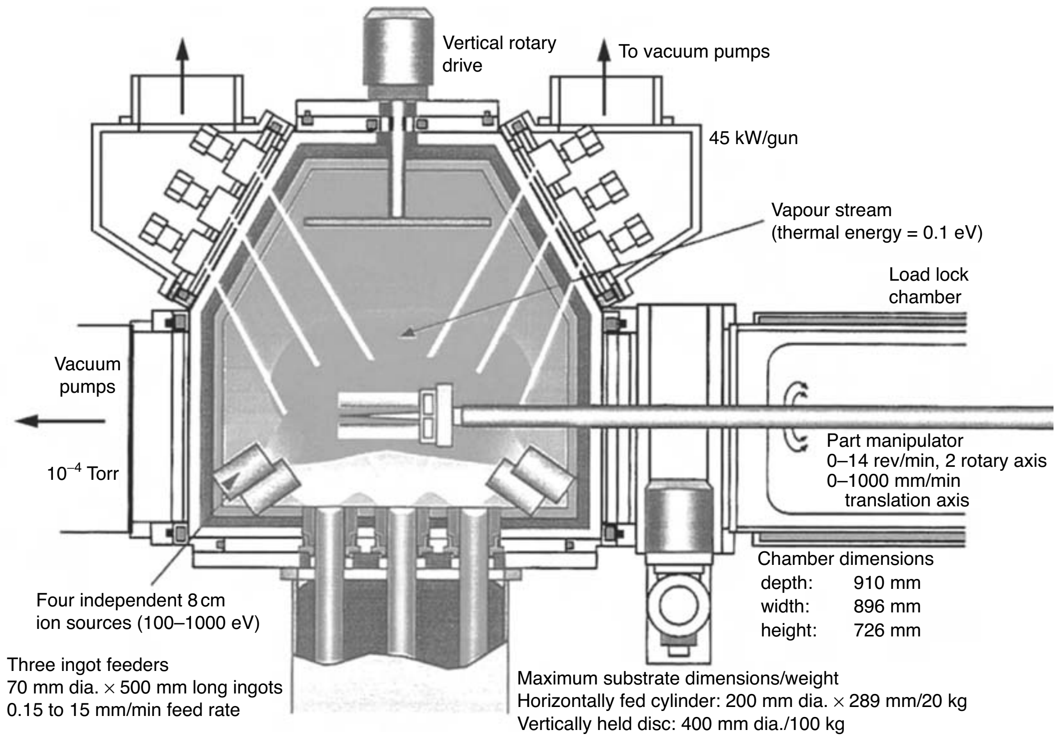


Fig. 5.2. Schematic illustration of the arrangement used for the electron beam physical vapour deposition (EB-PVD) method [15].

of the substrate. Figure 5.4 shows the microstructure of a typical 7 wt%Y_2O_3–ZrO_2 thermal barrier coating produced by the EB-PVD process, which has been shown to grow with a strong ⟨111⟩ texture [16]. The boundaries between the columnar deposits have been shown to be poorly bonded [17] – in effect, a bundle of columnar grains exists whose ends are attached to the substrate – so that a degree of strain tolerances is introduced into the coatings which would otherwise be prone to extreme brittleness and failure due to stresses developed during thermal cycling. Unfortunately, this also means that the coating quality is impaired if the surface to be coated is not perfectly clean; small imperfections are not covered up as they might be with other coating processes, but can result in a growth abnormality that is magnified as the coating thickens [7]. Experiments have shown that the size and morphology depends not just upon the substrate temperature, but also on the rate of component rotation which is required due to the complicated shape of the turbine aerofoils – EB-PVD being a line-of-sight process. At low temperatures and low rotational speeds, the columns vary not only in size from root to top, but also from one to another; see Figure 5.5. Increasing both the substrate temperature and the rotational speed improves the regularity and the parallelity of the microstructure and enlarges the column diameter. The need for component temperatures to be raised in this way prior to coating places further demands on the coating apparatus. For this purpose, either conventional radiation heating in the form of graphite furnaces placed in a preheating chamber, or else pre-heating by the electron beam itself is used.

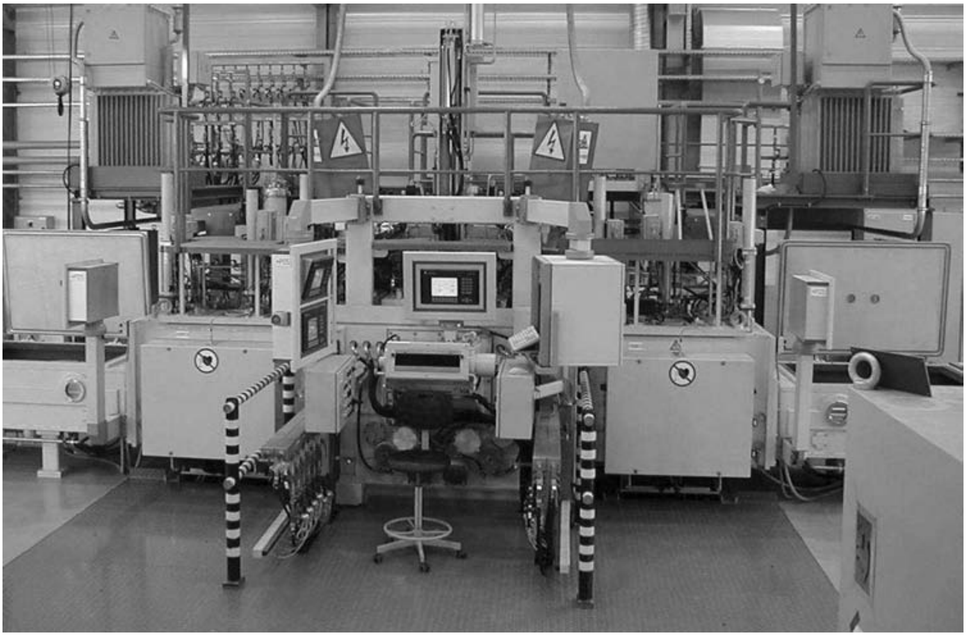


Fig. 5.3. Industrial-scale electron beam physical vapour deposition (EB-PVD) unit for the deposition of ceramic coatings on gas turbine components. (Courtesy of Matt Mede, ALD Vacuum Technologies.)

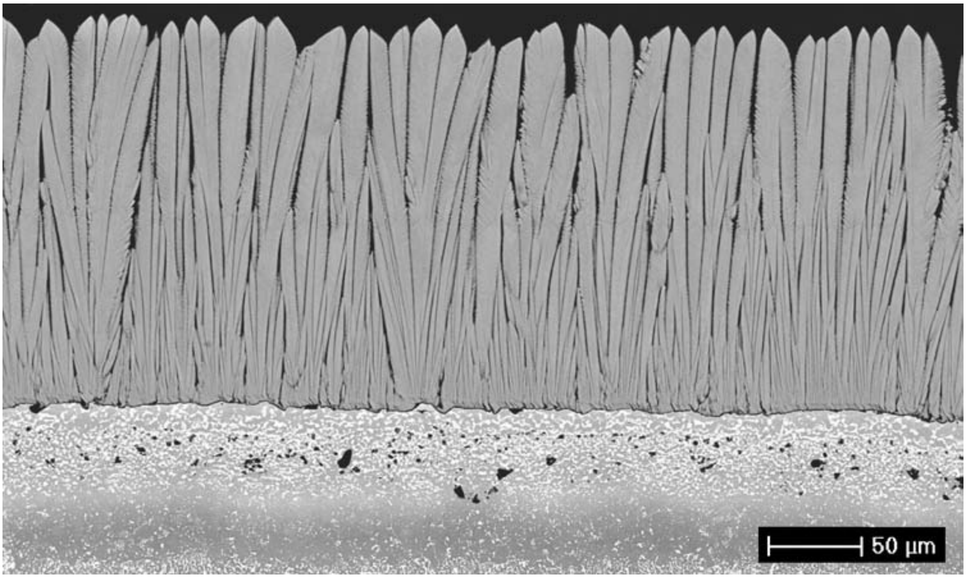


Fig. 5.4. Typical yttria-stabilised zirconia (YSZ) coating microstructure produced by the EB-PVD process. (Courtesy of Martijn Koolloos, NLR, the Netherlands.)

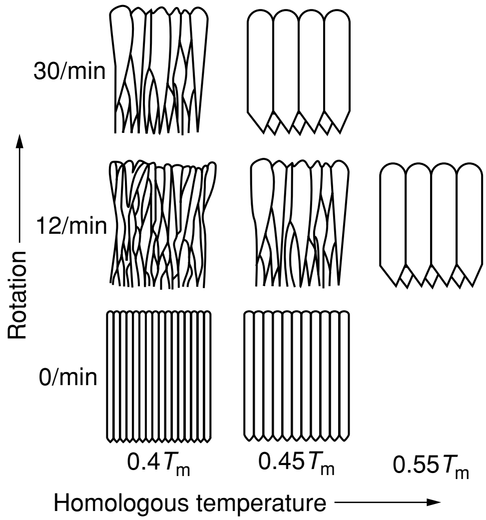


Fig. 5.5. Schematic illustration of the EB-PVD coating morphologies produced in YSZ material, as a function of substrate temperature and substrate revolution speed. (Courtesy of Uwe Schulz, DLR, Germany.)

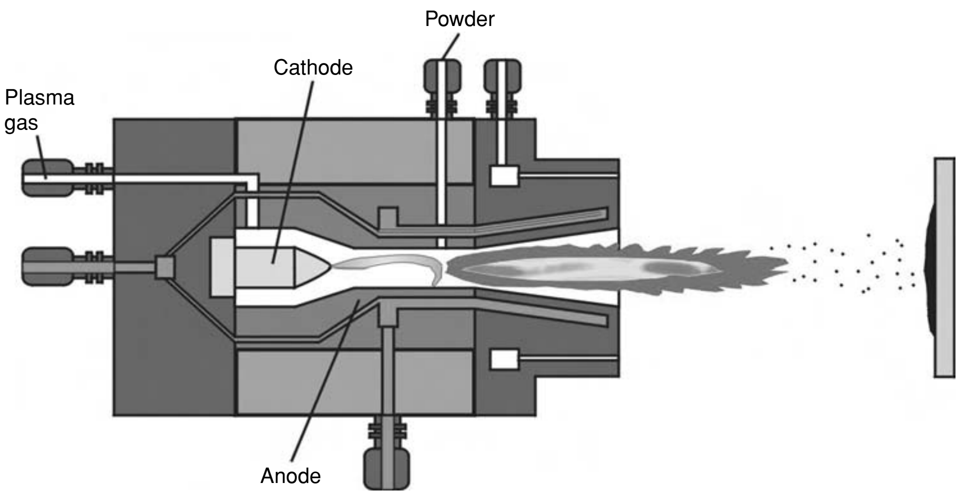


Fig. 5.6. Schematic illustration of the cross-section of a plasma torch in which the powder is injected internally.

5.1.2 *Plasma spraying*

Plasma spray processes utilise the energy contained in a thermally ionised gas to melt and propel fine metal or oxide particles, initially in powder form, to a surface such that they adhere and agglomerate to produce a coating [18–20]. The plasma itself consists of gaseous ions, free electrons and neutral atoms – with its temperature in the torch exceeding 10 000 °C. The elements of a typical plasma torch are shown in Figure 5.6. A gas, most typically argon or nitrogen, is made to flow around a tungsten cathode in an annular space housed by a water-cooled copper anode of complex shape. A high-frequency electrical

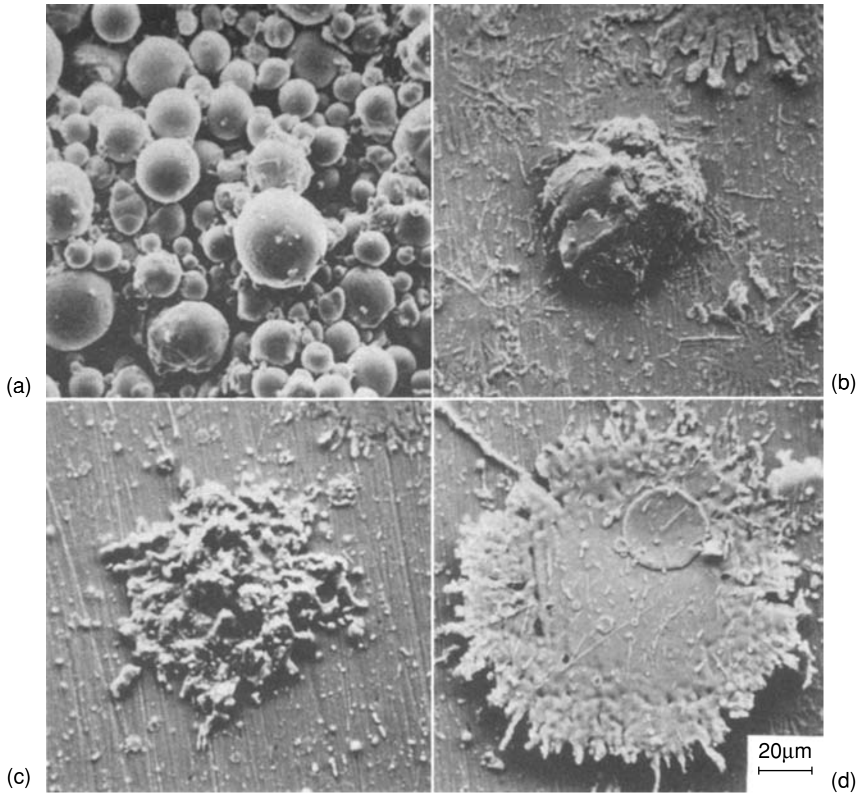


Fig. 5.7. Solidified NiCoCrAlY particles produced by plasma spraying: (a) initial condition, prior to spraying (b) and (c) partially melted, and (d) liquid condition upon impact [21].

discharge is then used to strike a direct current arc between the electrodes which is carried by the ionised plasma – causing a region of very high temperature into which particles of the powder to be sprayed are introduced. These are rapidly melted and then propelled towards the substrate to be coated. The entry of the powder should be introduced uniformily, either where the nozzle diverges or just beyond the exit of the torch. The powder velocities which are produced are very high – several hundred metres per second is common. The plasma-sprayed coatings are then built upon particle-by-particle, each producing a characteristic 'splat' morphology [21]; see Figure 5.7. Due to the large differences in thermal mass of particles and substrate, high cooling rates are achieved in the range 10^6 to 10^8 K/s. Most powder used for plasma spraying has a diameter between 5 and 60 μm; to achieve uniform heating and acceleration of it such that a high coating quality is produced, it should also be introduced into the plasma at a uniform rate. A narrow size range is preferred since this improves deposition efficiency; finer particles give denser deposits with less porosity, but with higher levels of residual stresses and oxide inclusions formed by oxidation of metallic particles during flight. A significant advantage of plasma spraying is that the composition of the powder usually closely matches that of the coating – which is not always the case

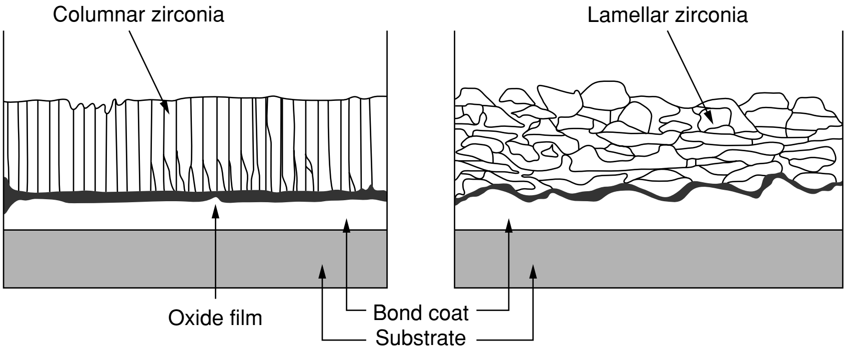


Fig. 5.8. Comparison of the microstructure of EB-PVD and plasma-sprayed TBCs [22].

with the EB-PVD process, due to different evaporation rates of the elements in the charge. A disadvantage of the process is that it is difficult to produce fully dense coatings without some porosity. Figure 5.8 compares the coating microstructure of plasma-sprayed and EB-PVD coatings, which display very different characteristics by virtue of the differing growth modes [22].

There are many types of plasma spraying apparatus available, from the basic systems which operate in air to the more sophisticated, fully automated systems which operate in a controlled environment or else in vacuum. With air plasma spraying (APS), the sprayed particles impact and deform, but perfect bonding is not achieved due to the presence of surface oxide films that cause oxide stringers to be present; consequently, they are characterised by their relatively low bond strengths and appreciable porosity. The low-pressure plasma spraying (LPPS) or vacuum plasma spraying (VPS) process circumvents some of these difficulties, because clean coatings are produced with virtually no oxide inclusions; see Figure 5.9. Since the plasma jet exhausts into a enclosed chamber held between 0.04 and 0.4 atm, significant expansions of the temperature isotherms and velocity contours of the plsama jet are achieved relative to APS; this is due to the increased mean free path between ions and electrons in the plasma – leading to very long spray distances. For example, spray distances for MCrAlX-type overlay coatings using the APS process are in the range 75 to 100 mm, whereas for VPS these are extended to greater than 400 mm [6]. At these so-called stand-off distances, the plume diameters are about 10 mm for APS and 50 mm for VPS. These characteristics lead to more uniform and consistent coatings, high densities close to the theoretical values, low residual stresses and the attainment of increased deposition thicknesses. Nevertheless, the use of VPS does come with some additional complications. The enclosed environment of the chamber necessitates robotic manipulation of the spray gun, vacuum interlocks, exhaust cooling, dust filtration and chamber-wall cooling, all of which add to the cost of the process. Consequently, the use of VPS is often used only where absolutely necessary; for combustor applications, for example, it is common to deposit an MCrAlX-type overlay bond coat using VPS prior to a YSZ-based thermal barrier coating using APS, to take advantage of lower cost and higher coating porosity of the latter process.

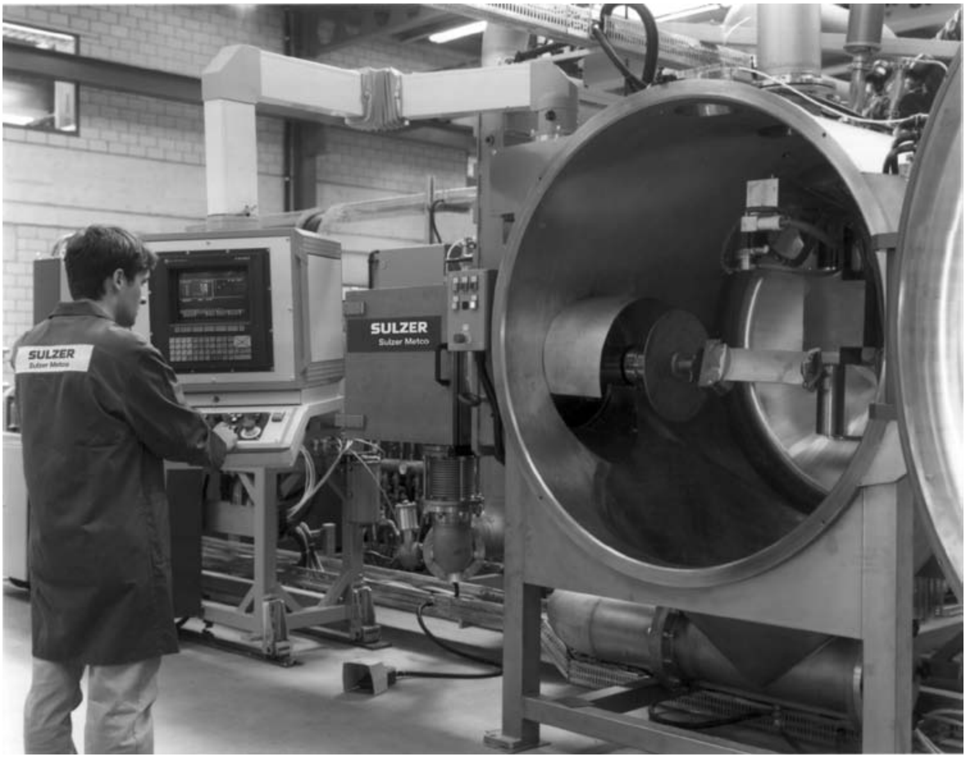


Fig. 5.9. Commercial vacuum plasma spraying (VPS) equipment for the production of thermally sprayed coatings. (Courtesy of Sulzer Metco.)

Recently, attention has been paid to techniques by which ‘segmentation cracks’ – running perpendicular to the coating surface and penetrating a significant fraction of the coating thickness – can be deliberately introduced into plasma-sprayed TBCs of YSZ [23]. The microstructure then resembles that of a TBC produced by the EB-PVD method; see Figure 5.10. It has been demonstrated that coatings fabricated in this way have improved resistance to spallation during thermal cycling, so that thicker TBCs can be produced which provide greater thermal insulation. Control of the substrate interpass temperature during spraying has been shown to be vital if segmentation cracks are to be introduced successfully – the range 500 to 800 °C is appropriate, with higher temperatures producing a greater crack density. The use of solid powder (fused and crushed) feedstock rather than the hollow-sphere (HOSP) variety is also helpful [24], although the latter gives rise to a greater porosity levels, which promote lower thermal conductivities. Although more research is required to elucidate the formation mechanism of segmentation cracks, it seems probable that the higher interpass temperatures cause successive splat layers to be present in the partially molten state, as evidenced by a common columnar solidification structure which then runs across them; the thermal strains induced by cooling are then able to promote segmentation cracking upon cooling, with improved intersplat bonding possibly playing a role. Unfortunately, the

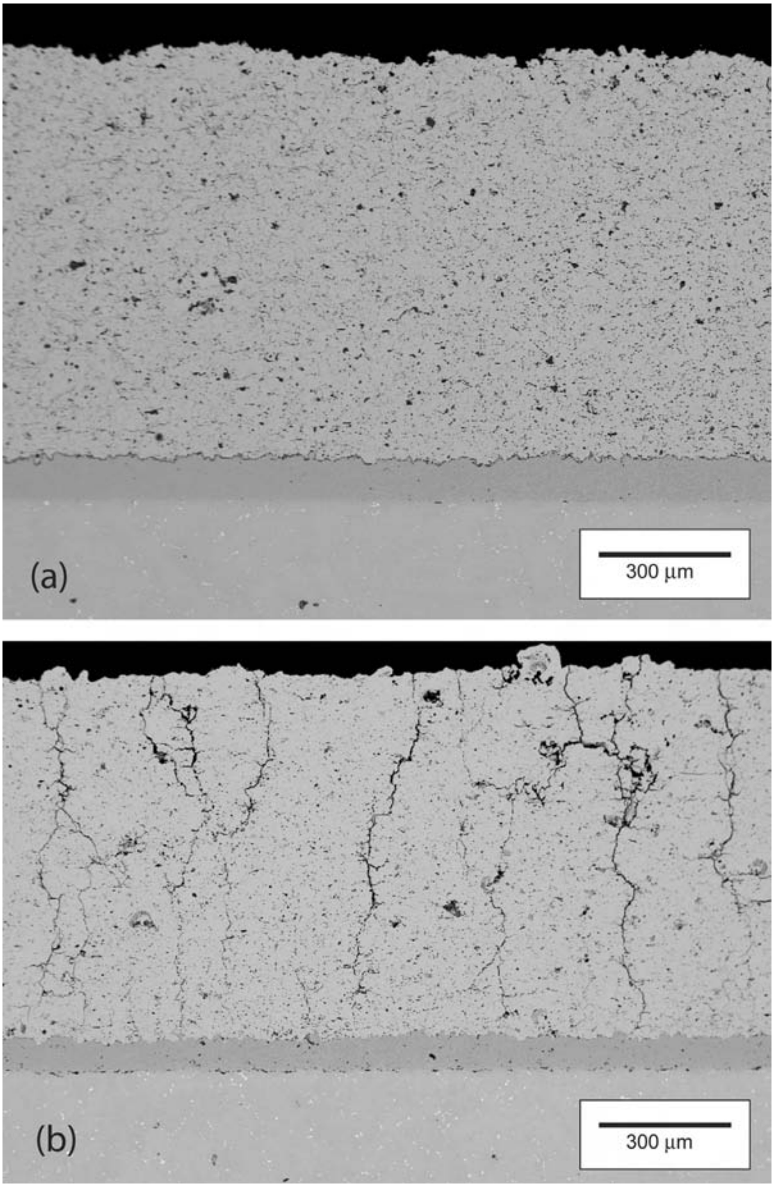


Fig. 5.10. Typical coating structures produced by plasma spraying of yttria-stabilised zirconia: (a) conventional processing conditions and (b) conditions designed to produce segmentation cracking. (Courtesy of Hongbo Guo.)

high porosity levels required for coatings of the lowest thermal conductivity are produced with low rather than high substrate temperatures – this indicates that, if segmentation cracks are to be introduced into any given coating material to improve its strain tolerance, then a greater thermal conductivity must be accepted.

5.1.3 Pack cementation and chemical vapour deposition methods

In the pack cementation process, the components to be coated are immersed in a powder mixture within a sealed or semi-sealed retort – this is then heated in a protective atmosphere until a coating of sufficient thickness is formed [25–28]. The pack consists of (i) the coating element source, (ii) an activator (such as NaF, NaCl or NH_4Cl) and (iii) an inert filler material, often alumina, which prevents the source from sintering [25]. The pack processes include chromising and siliconising, but aluminising is probably the most important and widely used in the gas turbine industry; for example, the vast majority of aeroengine turbine blade aerofoils are aluminised in some way to improve their resistance to high-temperature oxidation or as a surface preparation prior to the deposition of a TBC. In this case, the source can be either pure aluminium or an aluminium alloy (for example, Cr–30 wt%Al) depending upon the aluminium activity required; it is the difference between the activity of Al in the source and the substrate surface which represents the driving force for coating deposition. Broadly speaking, the activator reacts with the aluminium source to form aluminium halides; these are transported to the surface of the substrate where they are deposited, allowing them to react with the alloy and thus release aluminium.

The flexibility arising from the choice of aluminium activity in the source allows two distinct types of coating microstructure to be built up [4,29]. When the Al activity is high, an external brittle layer of δ–Ni_2Al_3 forms at the substrate surface (see Figure 5.11); aluminium is then the main diffusing species and the coating grows inwards. The diffusion coefficient of Al in δ is such that the process can then be carried out at relatively low temperatures, for example, 870 °C for 20 h – hence the term 'low-temperature, high-activity' (LTHA) is applied in this case. Subsequent heat treatment – for example, 1100 °C for 1 h – is used to promote the diffusion of Al inwards and Ni outwards such that a surface layer of β–NiAl is formed on the γ/γ' substrate, with an 'interdiffusion zone' (IDZ) forming between the two (see Figure 5.12); microanalysis using energy dispersive spectroscopy confirms that the IDZ is rich in slow-diffusing elements such as Re, W and Ta between the two (see Figure 5.13). Alternatively, if a low-activity pack is used then an external β–NiAl layer forms immediately during aluminisation, predominantly by the outward diffusion of nickel [4]. Since a relatively high temperature of 1100 °C is required in this case to promote reasonable diffusion rates, it is referred to as the 'high-temperature, low-activity' (HTLA) version of the process. Often, platinum-modified aluminide coatings are produced by electroplating a thin layer of Pt onto the surface of the superalloy; to encourage its adherence, a grit-blasting treatment is often applied prior to Pt electrodeposition. The incorporation of Pt into the aluminide layer in this way has been shown to improve the long-term oxidation resistance [30,31]. Traditionally, this has been attributed [32–34] to the exclusion of refractory elements from the outer aluminide layer caused by the presence of Pt; however, recent work [35] has disputed this view, the beneficial effect of platinum being linked to the minimisation of void formation at the coating/oxide interface and a retention of the oxide scale.

One drawback of the pack-cementation process is that it can only be used effectively to coat those blade surfaces which may be brought into direct contact with the pack, so

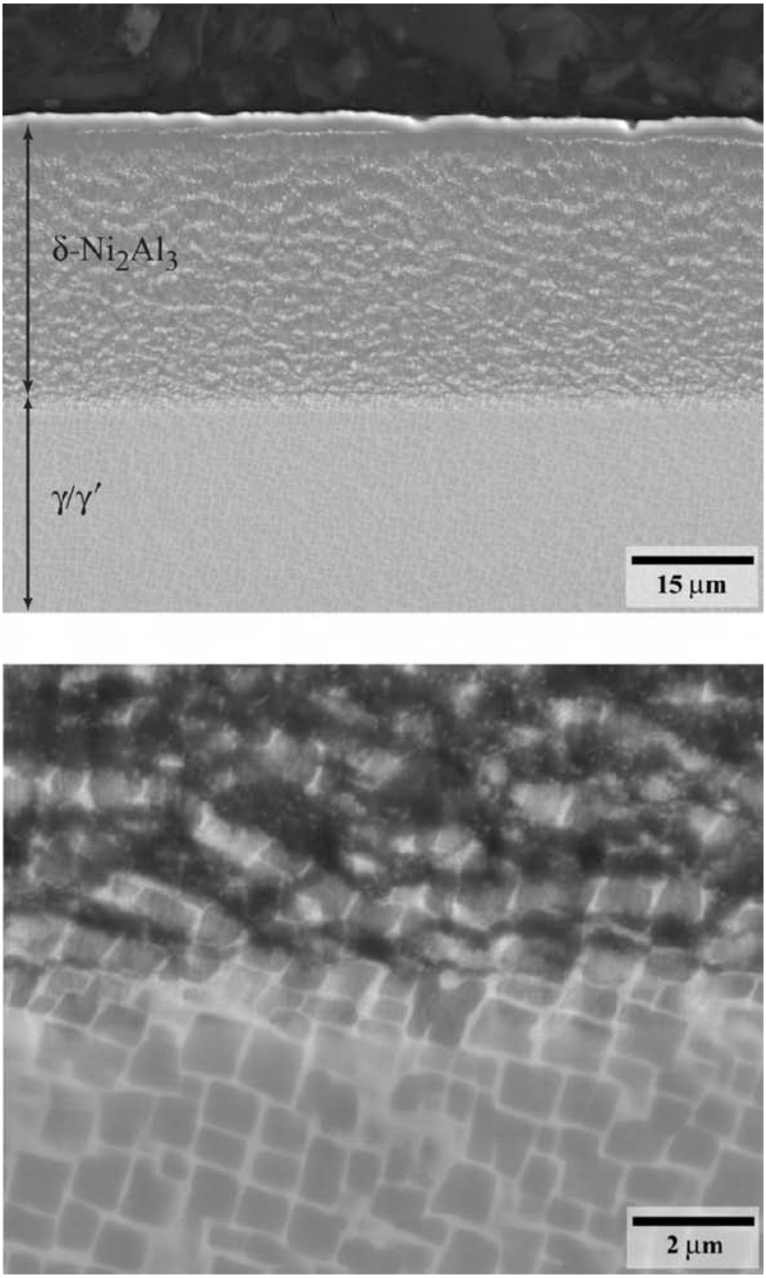


Fig. 5.11. Microstructures of the CMSX-10 single-crystal superalloy, following the low-temperature, high-activity, pack-cementation aluminisation process of 870 °C for 20 h. (Courtesy of Matthew Hook.)

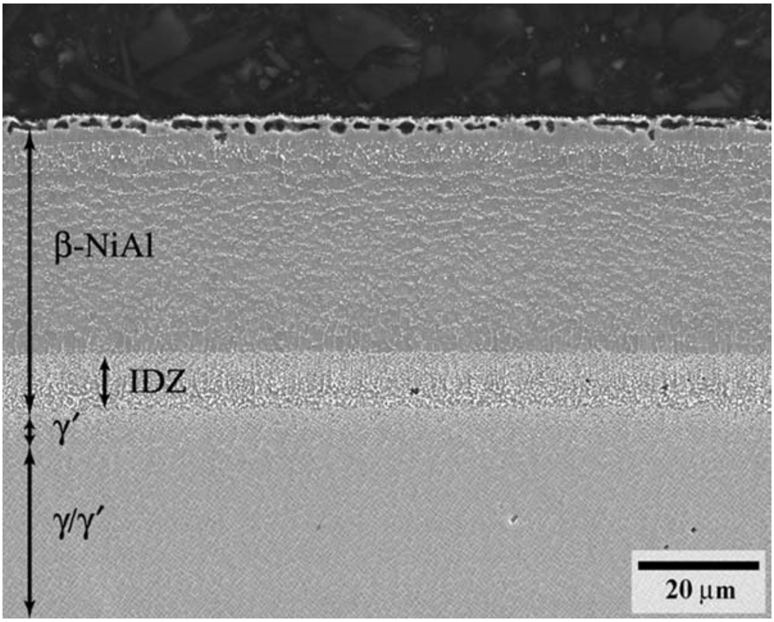
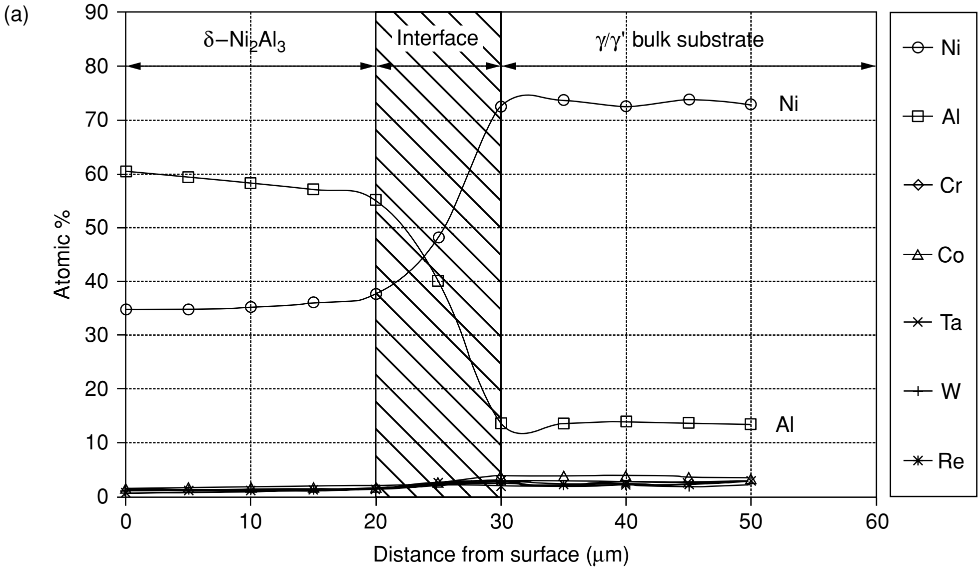


Fig. 5.12. The microstructure of the CMSX-10 superalloy following the low-temperature, high-activity pack-cementation aluminisation process, and subsequent heat treatment at 1100 °C for 1 h and further ageing at 875 °C for 16 h. (Courtesy of Matthew Hook.)

that the intricate cooling channels and internal surfaces found in many turbine blades are left uncoated [36,37]. To circumvent this difficulty, chemical vapour deposition (CVD) methods can be used such that aluminisation takes places exclusively in the gas phase [38]. Two distinct variants of gas-phase aluminisation can be identified. In the 'above-pack' process, the components are suspended within a reactor above beds of aluminium–chromium chips, upon which a powder of $AlCl_3$ is distributed. The reactor is filled with argon by means of a continuous gas feed, heated to a temperature of ~1100 °C, at which point the aluminisation reaction is initiated by the introduction of hydrogen gas into the Ar feed. The reactions which occur are beyond the scope of this book [36], but hydrogen plays a role in forming the reduced form of aluminium chloride, aluminium monochloride (AlCl), which reacts at the superalloy surface to produce the coating layer. The resulting reactants are transported to the surface of superalloy components by the thermal eddy currents which are set up within the reactor. This 'above-pack' process differs from the 'true' chemical vapour deposition method [36,37], in which the trichloride gas is generated outside of the reactor prior to its introduction into the reaction vessel – this allows its flow rate and activity to be accurately controlled, so that consistent and uniform aluminide layers are produced. A potential advantage of this approach, which needs further technological development, is that the generation and supply of reactant species outside of the reactor permits a possible 'reactive element co-deposition' so that additional elemental species may be incorporated into the coating to improve its effectiveness [39]. Schematic diagrams of the 'above-pack' and 'true' vapour deposition processes are given in Figure 5.14.



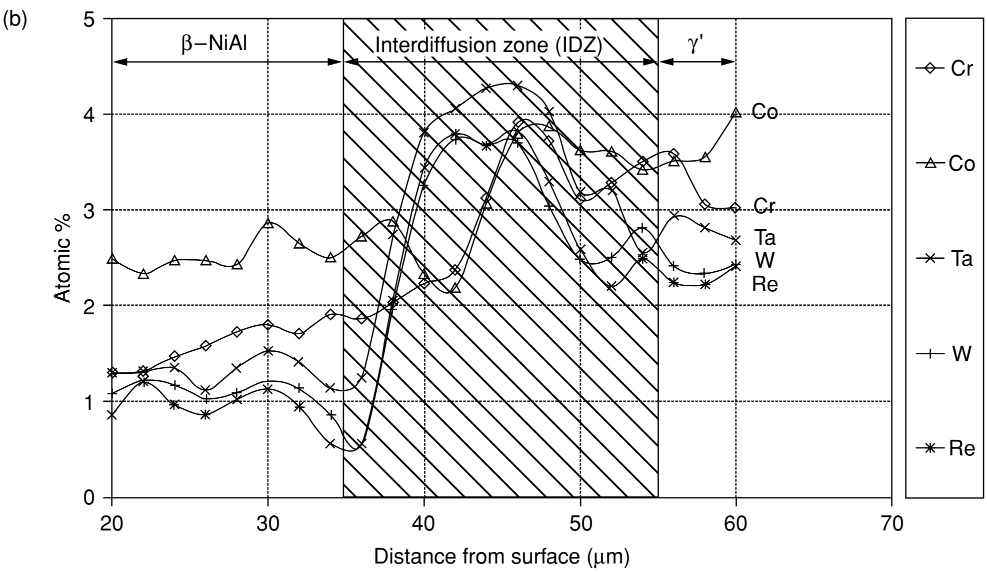


Fig. 5.13. Surface concentration profiles of the CMSX-10 superalloy (a) after the LTHA aluminisation process of 870 °C for 20 h, and (b) after subsequent heat treatment 1100 °C for 1 h and further ageing at 875 °C for 16 h. (Courtesy of Matthew Hook.)

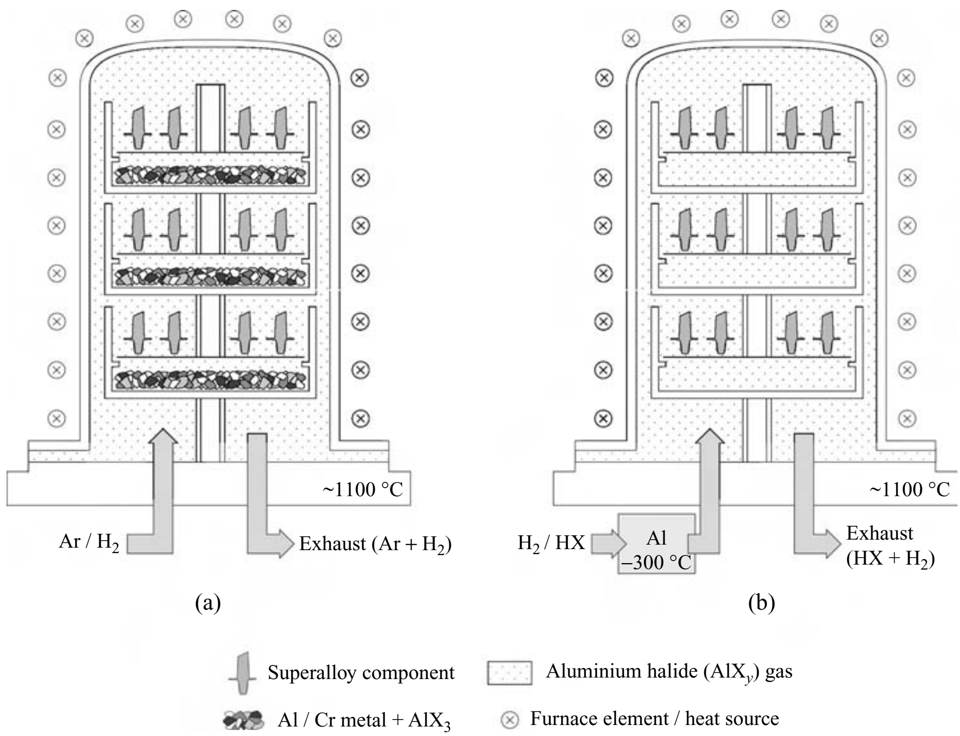


Fig. 5.14. Schematic diagram of the two reactor types employed for CVD aluminisation. (a) The 'above-pack' technique relies on the *in situ* evolution of the aluminium halide species, whereas the 'true' CVD process, (b), generates the aluminide halides externally and prior to introduction into the reactor.

5.2 Thermal barrier coatings

5.2.1 *Quantification of the insulating effect*

The potency of the insulation effect produced by a thermal barrier coating is best illustrated with the following quantitative example.

Example calculation

Consider a hollow, high-pressure turbine blade operating in a gas stream at 1450 °C, such that the outer and inner surfaces are at 1000 °C and 800 °C, respectively – the mean metal temperature is then 900 °C. The wall thickness can be taken as 5 mm; see Figure 5.15(a). Determine the decrease in metal temperature and steady-state heat flux arising from the addition of a 1 mm-thick thermal barrier coating (TBC); see Figure 5.15(b). The superalloy and TBC have thermal conductivities of 100 and 1 W/(m K), respectively. Assume that the heat transfer coefficient at the outer surface of the blade is unaltered by the addition of the TBC.

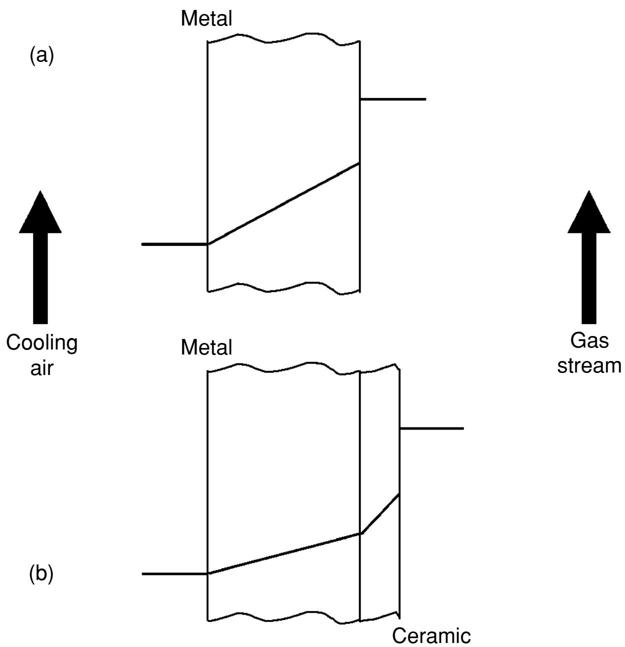


Fig. 5.15. Temperature profiles across a cooled turbine blade aerofoil (a) without a thermal barrier coating (TBC) and (b) with the TBC in place.

Solution Without the TBC, the temperature gradient across the superalloy is $200/5 \times 10^{-3}$ or 4×10^4 K/m. Multiplying by the thermal conductivity gives a value of the heat flux, q, of 4×10^6 W/m^2; this must equal the product of the temperature drop across the gas stream/metal interface of 450 °C and the film transfer coefficient, h. An estimate of this last quantity is therefore $4 \times 10^6/450$ or 8.9×10^3 W/(m^2 K).

With the TBC added, let x and y be the new temperatures at the metal/ceramic interface and ceramic/gas stream interfaces, respectively. The equivalence of the new heat flux, q', through the coated system means that

$$q' = 100 \times (x - 800)/5 \times 10^{-3} = 1 \times (y - x)/1 \times 10^{-3} = 8.9 \times 10^3(1450 - y)$$

On solving these equations one finds that the temperature of the outer surface of the superalloy is now 828 °C, so that the mean metal temperature is reduced to 814 °C. The new flux, q', is then 5.6×10^5 W/m^2, i.e. 14% of the value without the TBC in place. The analysis can be used to confirm that both the flux and mean metal temperature decrease as the thickness of the ceramic layer is increased and its thermal conductivity is reduced.

5.2.2 The choice of ceramic material for a TBC

The above considerations confirm the usefulness of a thermal barrier coating and its dependence upon the thermal conductivity of the material employed. It is well known that ceramic compounds have low values of thermal conductivity, but which one is best for this

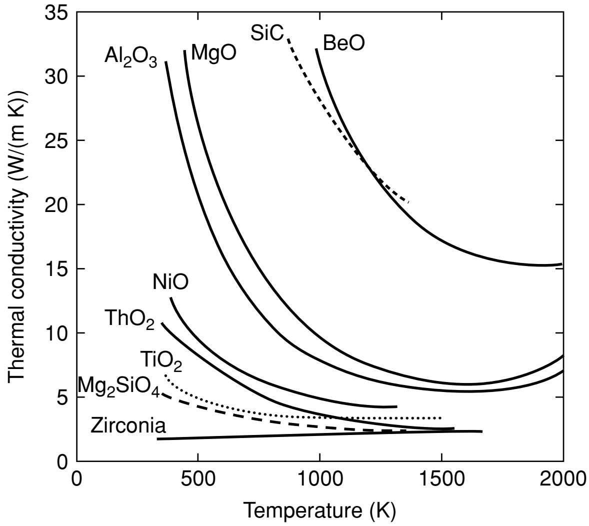


Fig. 5.16. The thermal conductivity of various polycrystalline oxides as a function of temperature [11,40].

application? Figure 5.16 presents some data [11,40] for thermally insulating oxide materials, in polycrystalline form. The thermal conductivities are in the range 1 to 30 W/(m K), the values decreasing with increasing temperature – the reasons for this are discussed in Section 5.2.3. Of the ceramic materials shown, zirconia (ZrO_2) displays the lowest conductivity, at about 2 W/(m K); moreover, no strong temperature dependence is found, so that this material does not suffer from the drawback of a steeply increasing conductivity at lower temperatures. These findings explain why ZrO_2-based ceramics have emerged as the materials of choice for TBC applications.

However, modern state-of-the-art TBCs are not fabricated from pure ZrO_2. Instead, they contain yttria (Y_2O_3) in the range 6 to 11% by weight, with a composition of 7 wt% of yttria being very commonly used [41]. This composition was originally identified by NASA during the 1970s as giving the best thermal cycle life in burner rig tests; see Figure 5.17 [42]. It turns out that this is the limiting composition at which the metastable 'non-transformable' tetragonal t′ phase is formed when the ceramic is quenched from the cubic phase field (see Figure 5.18). This avoids the tetragonal to monoclinic (t → m) phase transformation, which is associated with a 4% volume change and thus the poor thermal cycling resistance due to cracking and spalling at low Y_2O_3 contents; it also reduces the thermodynamic driving force for the dissociation at temperature into the equilibrium tetragonal t and cubic phases which inevitably occurs over time. The tetragonal t′ phase is thought to be the same tetragonal polymorph as the equilibrium tetragonal phase but with a very different microstructure, consisting of anti-phase domain boundaries and numerous twins [43]. When Y_2O_3 is combined with ZrO_2 in this way so that the t′ structure is formed, the resulting solution is known as 'partially stabilised zirconia'; the use of the adjective *partially* distinguishes it from 'fully stabilised zirconia' which arises at higher Y_2O_3 contents when the cubic phase is stable from ambient conditions to the melting point. For this reason, partially stabilised zirconia

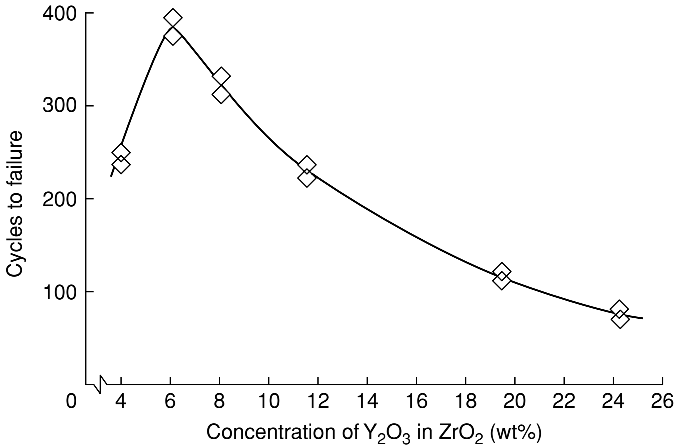


Fig. 5.17. Results from burner rig tests, which confirm the optimum yttria content of partially stabilised zirconia at about 7 wt% [42].

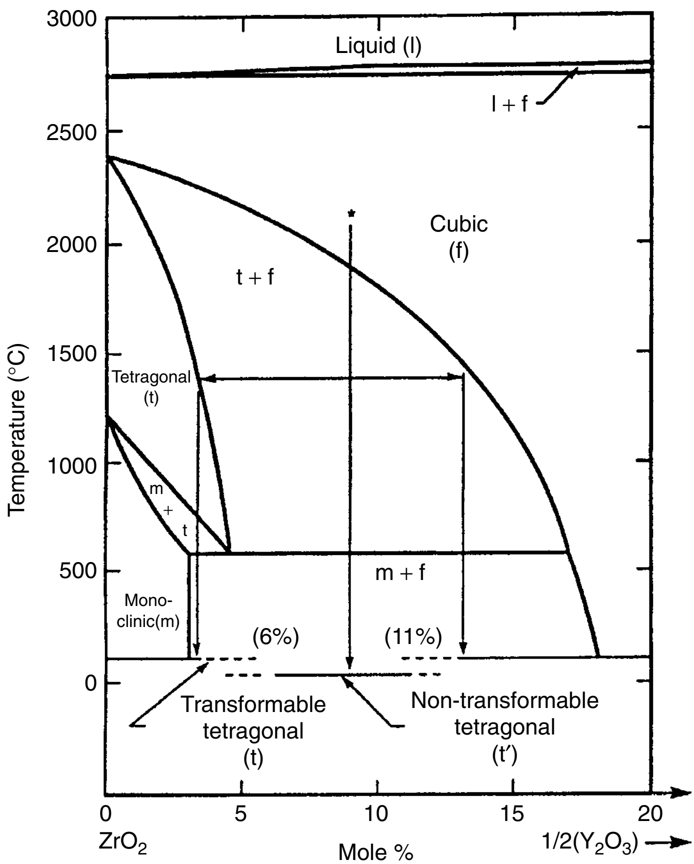


Fig. 5.18. Phase diagram for ZrO_2–Y_2O_3, showing the phase fields corresponding to the monoclinic (m), tetragonal (t) and cubic (f) phases, and the composition range for which the non-transformable tetragonal t′ form is found [44]. Note that sintering at 1400 °C of a composition corresponding to t′ must nonetheless lead to separation into the cubic and transformable tetragonal phases.

has emerged as the standard for TBC applications, being deposited in production onto combustor sections since the 1970s using thermal spraying and onto turbine nozzle guide vanes and blading since the 1980s using EB-VPD. It shows excellent thermal stability up to temperatures of 1200 °C, but unfortunately beyond this point it suffers from two drawbacks. First, partial reversion to the equilibrium t and cubic phases occurs with the tetragonal to monoclinic transformation then occurring upon subsequent thermal cycling [44], the sluggishness of this reverse transformation being due to the need for diffusion of Y through the zirconia. Second, extended exposure to high temperatures causes the sintering of the YSZ, with an associated increase in the elastic modulus [45]. The combination of these effects causes a loss of thermal cycling resistance.

Because of the importance of highly stable, low-thermal-conductivity ceramic materials for TBC applications, research has been carried out on compositions distinct from the 7YSZ standard. For example, CeO_2 rather than Y_2O_3 has been recommended (at a level of 25 wt%) for the stabilisation of ZrO_2 since no formation of the monoclinic phase was observed at 1400 °C and during limited exposures to 1600 °C; unfortunately, the erosion resistance is not improved by CeO_2 [46]. Similarly, 90% substitution of Sc_2O_3 for Y_2O_3 in so-called scandia/yttria-stabilised zirconia has been found to have significantly better tetragonal t′ phase stability at 1400 °C [47,48]. Alternatively, ZrO_2-based ceramics can be dispensed with entirely. For example, hafnia (HfO_2) is a good candidate for an alternative ceramic material; HfO_2–Y_2O_3 TBCs have been found to display thermal cycling resistance comparable to 7YSZ, but at higher Y_2O_3 compositions of up to 27 wt% when the crystal structure is fully cubic [49]. Furthermore, lanthanum zirconate ($La_2Zr_2O_7$), which displays the pyrochlore crystal structure, a thermal conductivity of about 1.6 W/(m K) and excellent thermal stability and thermal shock resistance, has been proposed as a TBC material [50,51], as have other rare-earth zirconates based upon Gd and Sm [52]; interestingly, none of these compounds contains structural vacancies, so there is some uncertainty as to the reason for the low thermal conductivity of these materials. Finally, lanthanum hexa-aluminate, which displays the magnetoplumbite structure, is considered by some to be a promising competitor to partially stabilised zirconia for operation above 1300 °C due to its resistance to sintering [53]. The science of these emerging ceramic materials for TBC applications has been reviewed recently in ref. [54]; at the time of writing, these new ceramics are not being used widely in service – although Pratt & Whitney claim to be using a $Gd_2Zr_2O_7$-based TBC for niche applications in military turbines. A significant difficulty is that many of the new ceramic materials have inadequate resistance to erosion which can occur due to sand ingestion and foreign object damage (FOD) accumulation. This is one further reason why partially stabilised rather than fully stabilised zirconia is preferred; in this system, the erosion resistance falls off monotonically with increasing yttria content.

5.2.3 *Factors controlling the thermal conductivity of a ceramic coating*

The choice of ceramic provides an intrinsic resistance to heat flow, but there are other features on the scale of the microstructure which provide a further *extrinsic* contribution. To appreciate this one should recognise that in crystalline solids in general, heat is transferred by

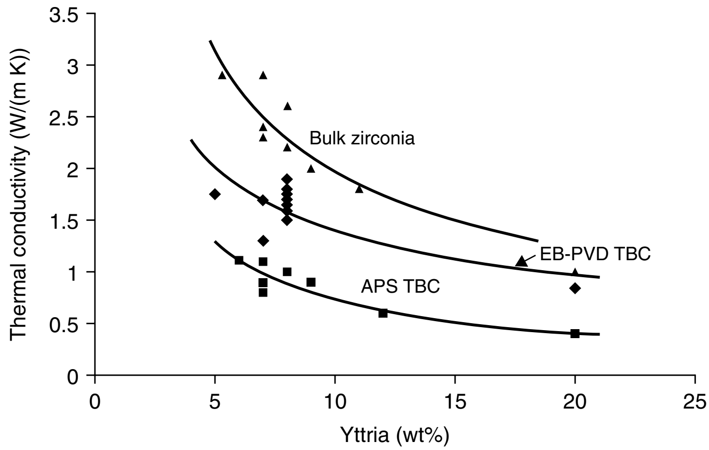


Fig. 5.19. The thermal conductivity of zirconia-based TBC materials as a function of yttria content and processing method [56].

three mechanisms [11]: (i) electrons, which are expected to provide the major contribution at very low temperatures, (ii) lattice vibrations, i.e. anharmonic phonon–phonon scattering, which will dominate at intermediate temperatures and (iii) radiation, i.e. photons, which will contribute strongly at very high temperatures. Ceramic materials in general are reasonable electronic insulators; indeed, zirconia is a very good one, with oxygen ion diffusion being responsible for electrical conductivity at elevated temperatures, so that any electronic contribution can be neglected. The data in Figure 5.16 support this conclusion, since the thermal conductivities decrease with increasing temperature, displaying in general the $1/T$ dependence characteristic of the Umklapp phonon–phonon process [55] – there is no increase with increasing temperature which would be expected from the electronic contribution. At very high temperatures the curves reach plateaux, becoming independent of temperature; as noted by Clarke [11], if not blocked, the radiative heat transport during thermal conductivity measurement can lead to an apparent increase in κ at high temperatures – there is evidence of this in Figure 5.16. Beyond 1250 °C, partially stabilised zirconia becomes significantly transparent to thermal radiation at wavelengths between 0.3 and 2.8 μm, with 90% of the incident radiation lying in this range with about 10% of the heat flux through the zirconia being due to radiation [56].

The above arguments indicate that further extrinsic reductions in thermal conductivity can be won by introducing inhomogeneities on the scale of the crystal lattice and grain structure. These considerations explain the significant decrease in the κ of yttria-stabilised zirconia (YSZ) observed with increasing yttria content and the potent influence of processing method used for deposition [56]; see Figure 5.19. Increasing the yttria content from 5 wt% to 20 wt% decreases the thermal conductivity by a factor of 2. However, the introduction of microstructural features such as intersplat boundaries and columnar grains in PS and EB-PVD coatings, respectively, has a comparable effect, with the former being more effective because they lie perpendicular to the direction of heat flow. Thus a coating has a lower